Linear Inversion
Bayesian inversion

Ketil Hokstad, February 2016
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Statistics recap

Probability density: $f(x)$

$$P(\hat{x} < x < \hat{x} + dx)$$

Expectation:

$$\begin{aligned} \mu &= E[x] = \int x f(x) dx \\ E[ax] &= a \int x f(x) dx = aE[x] \end{aligned}$$

Variance:

$$\begin{aligned} \sigma^2 = E[(x-\mu)^2] &= \int (x-\mu)^2 f(x) dx \\ E[a^2(x-\mu)^2] &= a^2 E[(x-\mu)^2] \end{aligned}$$

Gaussian (normal) distribution:

$$
\begin{aligned}
f(x) &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \\
x &\sim \mathcal{N}(\mu, \sigma)
\end{aligned}
$$

Standard normal distribution:

$$
\begin{aligned}
f(z) &= \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \\
z &\sim \mathcal{N}(0, 1)
\end{aligned}
$$

Standard variable

if $\quad x \sim \mathcal{N}(\mu, \sigma)$

then $\quad z = \dfrac{x - \mu}{\sigma} \sim \mathcal{N}(0, 1)$

Log normal distribution

if $\quad x \sim \mathcal{N}(\mu, \sigma)$

and $\quad x = \ln(y)$

then $\quad y$ is log normal

Standardization of variables:
Frequently used for «feature engineering» in machine learning

Multivariate distributions

$$
\begin{aligned}
f(\mathbf{x}) &= f(x_1, x_2, \cdots, x_N) \\
\mu_i &= E(x_i) = \int x_i f(\mathbf{x}) dx_i \\
\Sigma_{ij} &= \text{cov}(x_i, x_j) = E[(x_i - \mu_i)(x_j - \mu_j)]
\end{aligned}
$$

$$
\begin{aligned}
\mathbf{\Sigma}^T &= \mathbf{\Sigma} \\
\mathbf{x}^T \mathbf{\Sigma} \mathbf{x} &\geq 0
\end{aligned}
$$

Symmetric and positive definite

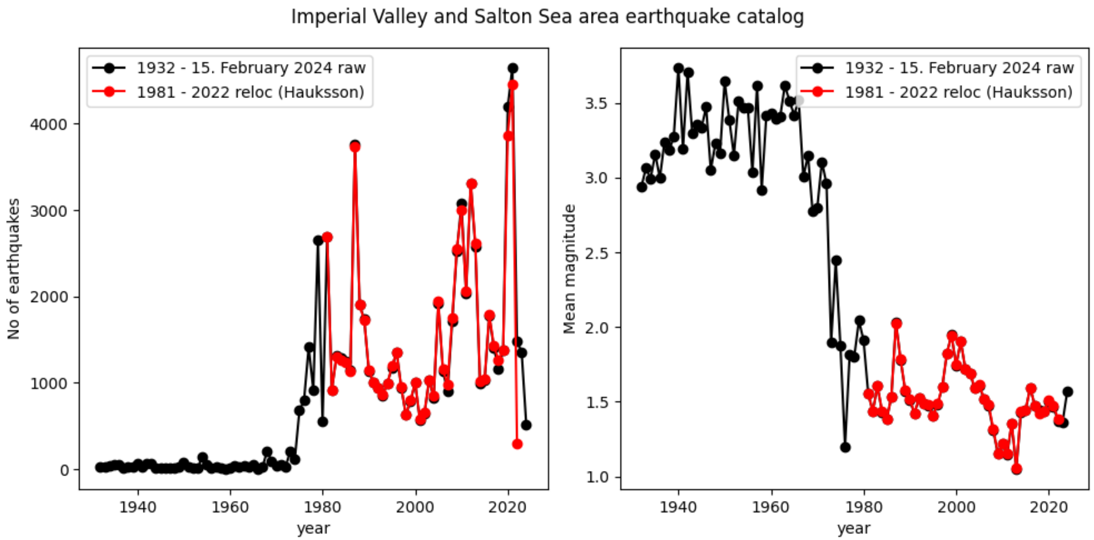
Correlation:

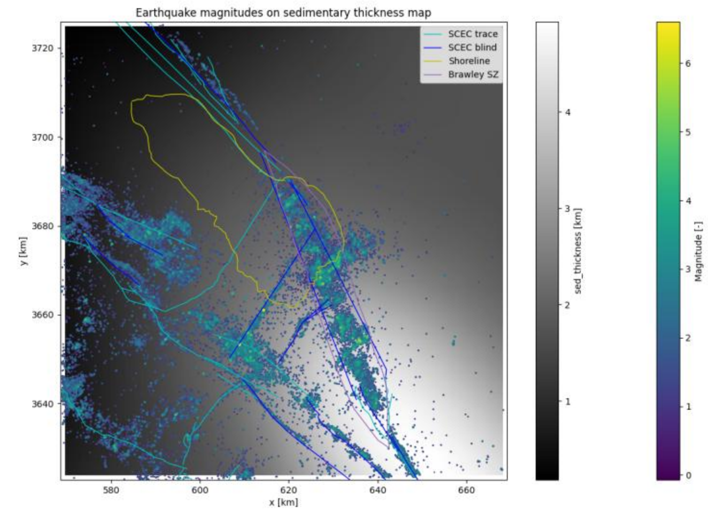
$$
\rho(x_i, x_j) = \frac{\text{cov}(x_i, x_j)}{\sqrt{\sigma_i^2 \sigma_j^2}}
$$

Independent random variables

$$
\begin{aligned}
f(\mathbf{x}) &= \prod_{i=1}^{N} f(x_i) \\
\text{cov}(x_i, x_j) &= 0
\end{aligned}
$$

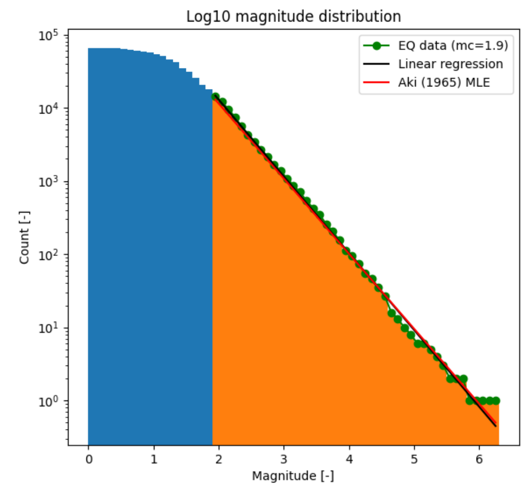
SCEC earthquake catalog 1932 – 15.feb2024





Geothermal industry starting ~1980

- More earthquakes
- Smaller magnitudes on average
- Can be detected due to denser network



Exponential laws and power laws are common in nature

Exponential laws:

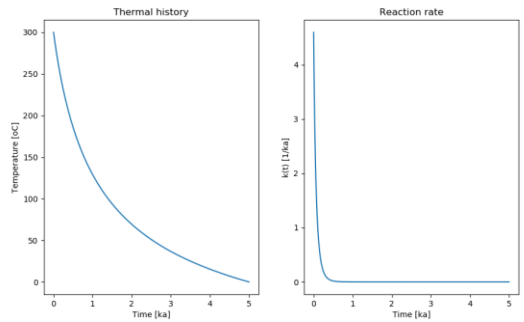
$$y = a^{-bx}$$

Linear in log-plots

$$\ln y = \ln a - bx$$

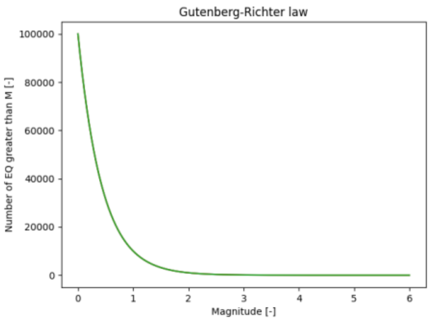
Example1: Arrhenius equation

$$k = Ae^{-\frac{E}{kT}}$$



Example2: Gutenberg-Richter

$$N_{\geq M} = 10^{a\,-bM}$$



Power laws:

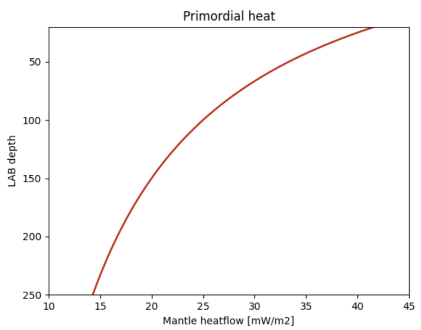
$$y = \frac{c}{x^n}$$

Linear in log-log plots

$$\ln y = \ln c - n \ln x$$

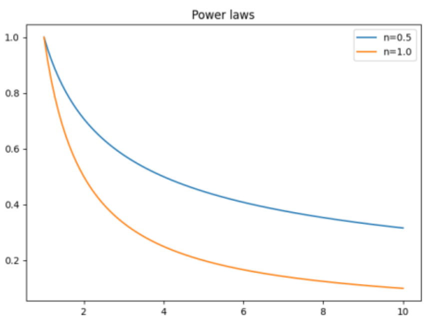
Example1: Primordial heat

$$Q_M = \frac{50}{1 + 0.01 z_{LAB}}$$



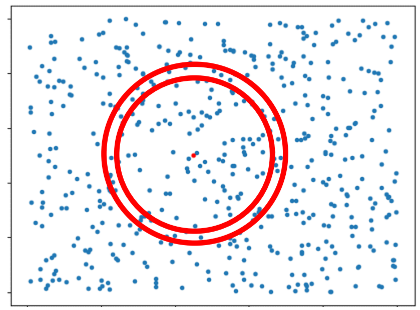
Example2: EQ correlations

$$\Gamma = \frac{C}{r^a}$$

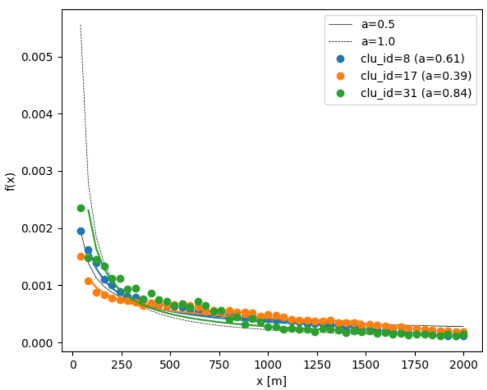


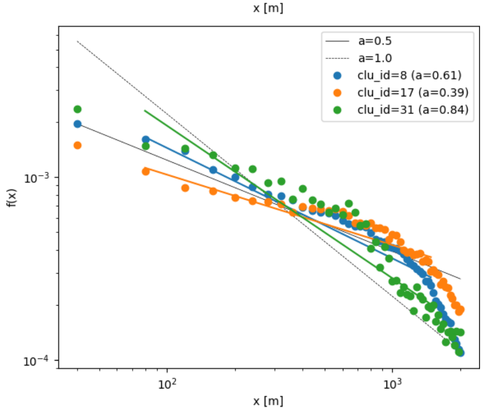
Two-point correlation

- Count events between r and $r + \Delta r$
- Normalize by $\Delta A = 2\pi r \Delta r$



Fun fact: The same two-point correlation methodology is used to estimate the density of galaxies in the universe

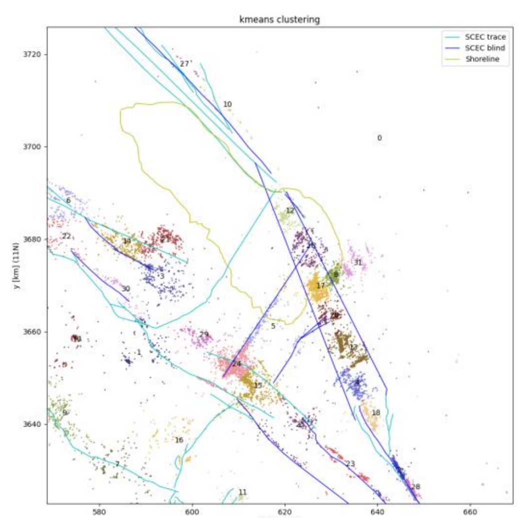




Power-law correlation:

$$\Gamma(r) \sim \frac{1}{r^a}$$

$a \sim 1/2$



SCEC catalog, kmeans clustering

Porosity:

Co-variance: $\boldsymbol{\Sigma}_{\psi} = \sigma_{\psi}^{2}\Gamma(r)$

pdf: $p(\Psi) \propto e^{-\frac{1}{2}[\Psi - \boldsymbol{\mu}_{\psi}]^{T}\boldsymbol{\Sigma}_{\psi}^{-1}[\Psi - \boldsymbol{\mu}_{\psi}]}$

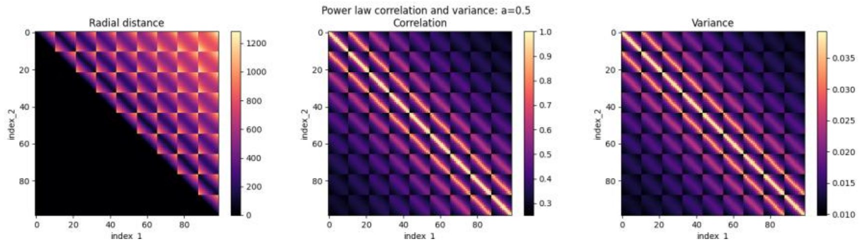
$$\boldsymbol{\Psi} = [\psi(\mathbf{x}_1) \quad \ldots \quad \psi(\mathbf{x}_n)]^{T}$$

log-normal: $\psi = \ln \phi$

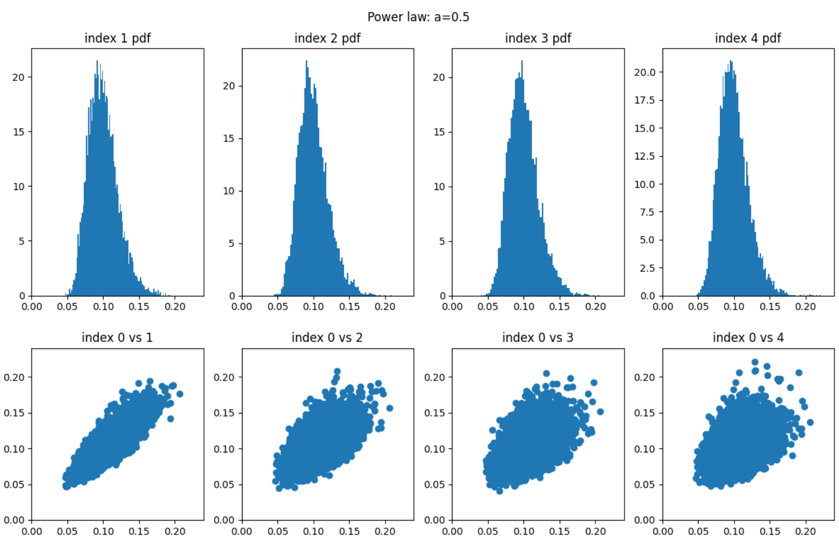
Permeability:

$$\ln k(\mathbf{x}_i) = \alpha\phi(\mathbf{x}_i) + \beta$$

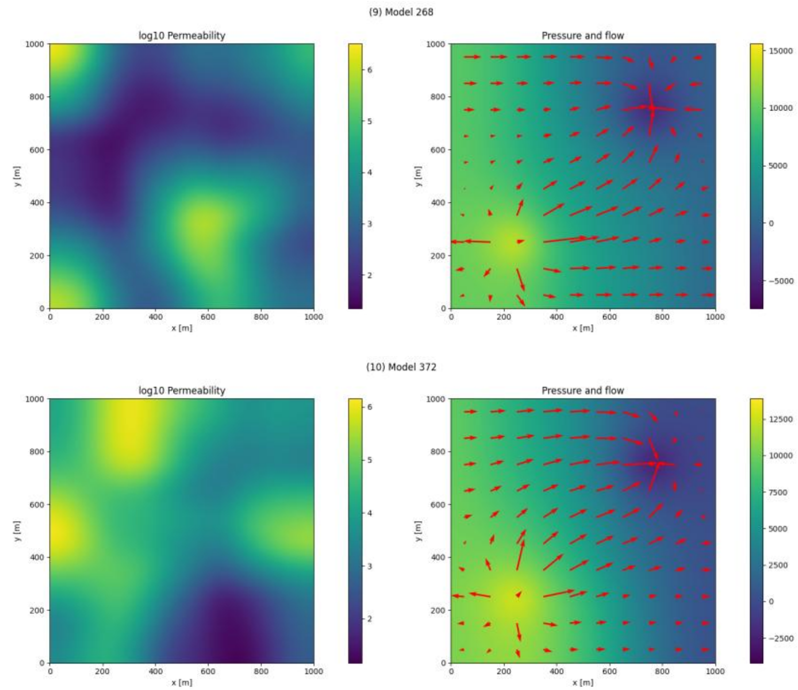
Sampling the porosity distribution: $a = 0.5$



Correlation and covariance matrices



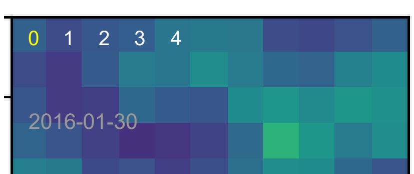
Some correlations and histogram distributions
10000 samples drawn from the multivariate distribution



Permeability

Fluid flow, injector/producer

Flow simultion with the PorePy
2 models (out of 10000)
Python code from U. of Bergen



Bayes theorem (Bayes rule)

Joint, conditional and marginal distributions:

$$
\begin{aligned}
f(\mathbf{x},\mathbf{y}) \quad &: \quad \text{joint} \\
f(\mathbf{x}|\mathbf{y}) \quad &: \quad \text{conditional} \\
f(\mathbf{x}) \quad &= \quad \int f(\mathbf{x},\mathbf{y})d\mathbf{y} \text{ : marginal}
\end{aligned}
$$

Thomas Bayes
(1701 - 1761)

Bayes theorem

$$
\begin{aligned}
f(\mathbf{x},\mathbf{y}) &= f(\mathbf{x}|\mathbf{y})f(\mathbf{y}) \\
f(\mathbf{x},\mathbf{y}) &= f(\mathbf{y}|\mathbf{x})f(\mathbf{x}) \\
f(\mathbf{x}|\mathbf{y})f(\mathbf{y}) &= f(\mathbf{y}|\mathbf{x})f(\mathbf{x})
\end{aligned}
$$

Geophysical inversion – Bayes theorem

Bayes theorem; posterior, likelihood, prior

$$\underbrace{f(\mathbf{m}|\mathbf{d})}_{\text{Posterior distribution}} \quad \propto \quad \underbrace{f(\mathbf{d}|\mathbf{m})}_{\text{Likelihood function}}\underbrace{f(\mathbf{m})}_{\text{Prior distribution}}$$

Likelihood function:

$$L(\mathbf{m}|\mathbf{d}) \quad = \quad f(\mathbf{d}|\mathbf{m})$$

$$\begin{aligned} \mathbf{d} \quad &= \quad \text{data (observations)} \\ \mathbf{m} \quad &= \quad \text{model} \end{aligned}$$

The likelihood function is determined by a <u>forward modeling operator:</u>

- Tomography equation
- Zoeppritz equation
- Wave equation
- Maxwell equation
- Magnetostatic equation
- Newton’s law of gravity
- Rock-physics relations

Linear, non-linear and linearized problems

Linear:

$$\mathbf{d} = \mathbf{A}\mathbf{m} + \mathbf{n}$$

Non-linear:

$$\mathbf{d} = \mathcal{F}(\mathbf{m}) + \mathbf{n}$$

Gaussian assumption:

$$\begin{aligned} \mathbf{n} &\sim \mathcal{N}(0, \mathbf{\Sigma}_e) \\ \mathbf{m} &\sim \mathcal{N}(\mu_m, \mathbf{\Sigma}_m) \end{aligned}$$

Linearized

$$\begin{aligned} \mathbf{d} &= \mathcal{F}(\mathbf{m}_0) + \mathbf{G}(\mathbf{m} - \mathbf{m}_0) + \mathbf{n} \\ \Delta\mathbf{d} &= \mathbf{G}\Delta\mathbf{m} + \mathbf{n} \end{aligned}$$

$$\begin{aligned} \Delta\mathbf{d} &= \mathbf{d} - \mathcal{F}(\mathbf{m}_0) \\ \Delta\mathbf{m} &= \mathbf{m} - \mathbf{m}_0 \\ \mathbf{G} &= \frac{\partial \mathcal{F}}{\partial \mathbf{m}}(\mathbf{m}_0) \end{aligned}$$

Types of inverse problems and solutions

- Two main types of inverse problems:
  - Linear, linearized
  - Non-linear

$$f(\mathbf{x}|\mathbf{y})f(\mathbf{y}) \quad = \quad f(\mathbf{y}|\mathbf{x})f(\mathbf{x})$$

- Two main types of solutions:
  - Optimization
    - ✓ Prior uncertainty
  - Statistical
    - ✓ Prior uncertainty
    - ✓ Posterior uncertainty
- Both types can be developed from a common statistical framework; Bayes rule

Likelihood - linear case

Likelihood function:

$$
\begin{aligned}
L(\mathbf{m}|\mathbf{d}) &\propto e^{-\frac{1}{2}(\mathbf{d}-\mathbf{A}\mathbf{m})^T \boldsymbol{\Sigma}_e^{-1}(\mathbf{d}-\mathbf{A}\mathbf{m})} \\
&= e^{-\frac{1}{2}[\mathbf{W}(\mathbf{d}-\mathbf{A}\mathbf{m})]^T[\mathbf{W}(\mathbf{d}-\mathbf{A}\mathbf{m})]}
\end{aligned}
$$

Data weights:

$$
\begin{aligned}
\mathbf{W}^T\mathbf{W} &= \boldsymbol{\Sigma}_e^{-1} \\
L(\mathbf{m}|\mathbf{d}) &= e^{-\frac{1}{2}(\hat{\mathbf{d}}-\hat{\mathbf{A}}\mathbf{m})^T(\hat{\mathbf{d}}-\hat{\mathbf{A}}\mathbf{m})}
\end{aligned}
$$

Weighted data:

$$
\hat{\mathbf{d}} = \mathbf{W}\mathbf{d}
$$

Least squares solution

Define objective function:

$$\Phi = \frac{1}{2}(\hat{\mathbf{d}} - \hat{\mathbf{A}}\mathbf{m})^T(\hat{\mathbf{d}} - \hat{\mathbf{A}}\mathbf{m})$$

Minimize $\Phi \Leftrightarrow$ maximize L:

$$\begin{aligned}
\frac{\partial \Phi}{\partial \mathbf{m}} &= 0 \\
\frac{\partial \Phi}{\partial \mathbf{m}} &= -\frac{1}{2}[\hat{\mathbf{A}}^T(\hat{\mathbf{d}} - \hat{\mathbf{A}}\mathbf{m}) + (\hat{\mathbf{d}} - \hat{\mathbf{A}}\mathbf{m})^T\hat{\mathbf{A}}] = 0
\end{aligned}$$

$$\begin{aligned}
\hat{\mathbf{A}}^T(\hat{\mathbf{d}} - \hat{\mathbf{A}}\mathbf{m}) &= 0 \\
\hat{\mathbf{A}}^T\hat{\mathbf{d}} - \hat{\mathbf{A}}^T\hat{\mathbf{A}}\mathbf{m} &= 0
\end{aligned}$$

Details - components

Objective function:

$$\Phi = \frac{1}{2}\sum_i \sum_j (d_i - A_{ij}m_j)(d_i - A_{ij}m_j)$$

Minimize:

$$\begin{aligned}\frac{\partial \Phi}{\partial m_k} &= -\sum_i \sum_j [(A_{ij}\delta_{jk})(d_i - A_{ij}m_j)] \\ &= -\sum_i \sum_j [A_{ik}(d_i - A_{ij}m_j)] = 0\end{aligned}$$

This is the kth equation of

$$\hat{\mathbf{A}}^T\hat{\mathbf{d}} - \hat{\mathbf{A}}^T\hat{\mathbf{A}}\mathbf{m} = 0$$

Least-squares (MLH) solution:

$$\mathbf{m} = \underbrace{[\hat{\mathbf{A}}^T\hat{\mathbf{A}}]^{-1}\hat{\mathbf{A}}^T}_{\text{pseudo inverse}}\hat{\mathbf{d}}$$

Re-introduce $\mathbf{W}$ (if you want)

$$\begin{aligned}
\mathbf{m} &= [(\mathbf{W}\mathbf{A})^T(\mathbf{W}\mathbf{A})]^{-1}(\mathbf{W}\mathbf{A})^T(\mathbf{W}\mathbf{d}) \\
&= [\mathbf{A}^T\mathbf{W}^T\mathbf{W}\mathbf{A})]^{-1}\mathbf{A}^T\mathbf{W}^T\mathbf{W}\mathbf{d} \\
&= [\mathbf{A}^T\mathbf{\Sigma}_e^{-1}\mathbf{A})]^{-1}\mathbf{A}^T\mathbf{\Sigma}_e^{-1}\mathbf{d}
\end{aligned}$$

So far we have only looked at the MLH term;
next we will introduce priors

Posterior solution and prior

Posterior distribution:

$$\begin{aligned} f(\mathbf{m}|\mathbf{d}) &\propto f(\mathbf{d}|\mathbf{m})f(\mathbf{m}) && \text{(Posterior pdf)} \\ f(\mathbf{d}|\mathbf{m}) &\propto e^{-\frac{1}{2}(\mathbf{d}-\mathbf{A}\mathbf{m})^T\mathbf{\Sigma}_e^{-1}(\mathbf{d}-\mathbf{A}\mathbf{m})} && \text{(Likelihood function)} \\ f(\mathbf{m}) &\propto e^{-\frac{1}{2}(\mathbf{m}-\mathbf{m}_0)^T\mathbf{\Sigma}_m^{-1}(\mathbf{m}-\mathbf{m}_0)} && \text{(Prior pdf)} \end{aligned}$$

Define objective function motivated by the posterior

$$\Phi = \frac{1}{2}[(\mathbf{d}-\mathbf{A}\mathbf{m})^T\mathbf{\Sigma}_e^{-1}(\mathbf{d}-\mathbf{A}\mathbf{m}) + (\mathbf{m}-\mathbf{m}_0)^T\mathbf{\Sigma}_m^{-1}(\mathbf{m}-\mathbf{m}_0)]$$

Minimize objective function Φ ⇔ maximize probability density $f(\mathbf{m}|\mathbf{d})$

Minimize $\Phi \Leftrightarrow$ maximize $f(\mathbf{m}|\mathbf{d})$:

$$\frac{\partial \Phi}{\partial \mathbf{m}} = 0$$

MLH part is as before; new prior terms:

$$-\mathbf{A}^T \boldsymbol{\Sigma}_e^{-1} \mathbf{d} + \mathbf{A}^T \boldsymbol{\Sigma}_e^{-1} \mathbf{A} \mathbf{m} + \boldsymbol{\Sigma}_m^{-1} (\mathbf{m} - \mathbf{m}_0) = 0$$

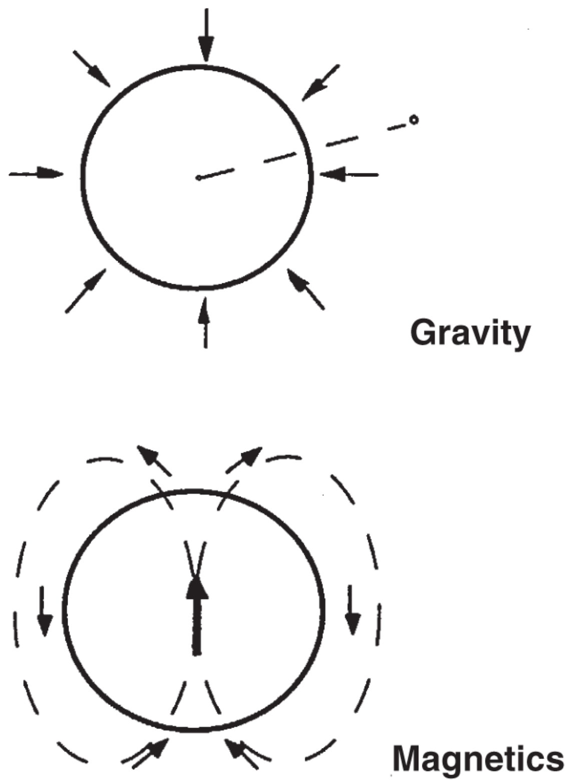
Get the unknown $\mathbf{m}$ on the l.h.s.

$$[\mathbf{A}^T \boldsymbol{\Sigma}_e^{-1} \mathbf{A} + \boldsymbol{\Sigma}_m^{-1}] \mathbf{m} = \mathbf{A}^T \boldsymbol{\Sigma}_e^{-1} \mathbf{d} + \boldsymbol{\Sigma}_m^{-1} \mathbf{m}_0$$

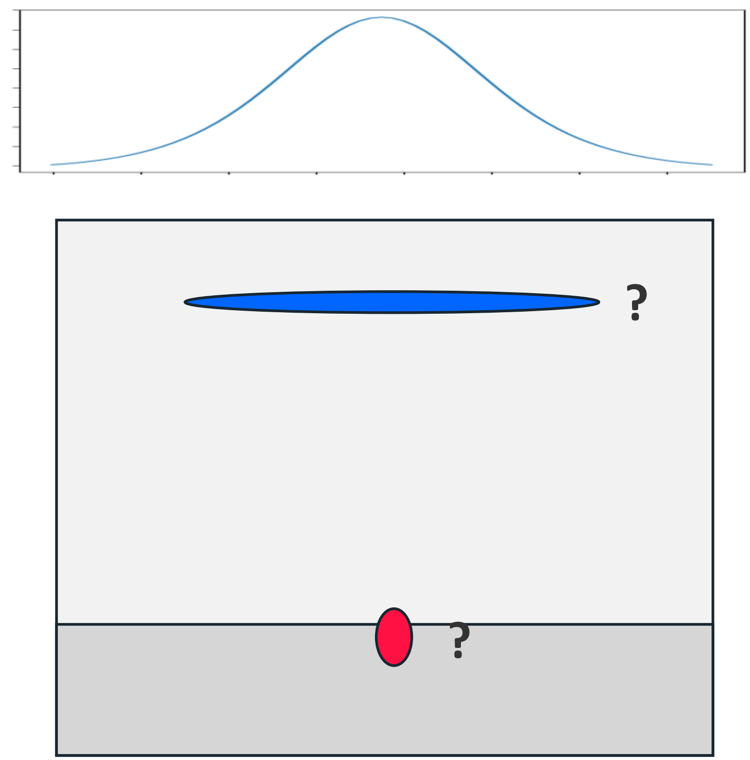
Regularized pseudo inverse:

$$\mathbf{m} = [\mathbf{A}^T \boldsymbol{\Sigma}_e^{-1} \mathbf{A} + \boldsymbol{\Sigma}_m^{-1}]^{-1} [\mathbf{A}^T \boldsymbol{\Sigma}_e^{-1} \mathbf{d} + \boldsymbol{\Sigma}_m^{-1} \mathbf{m}_0]$$

Linear inversion – gravity and magnetics



Big and shallow or small and deep?



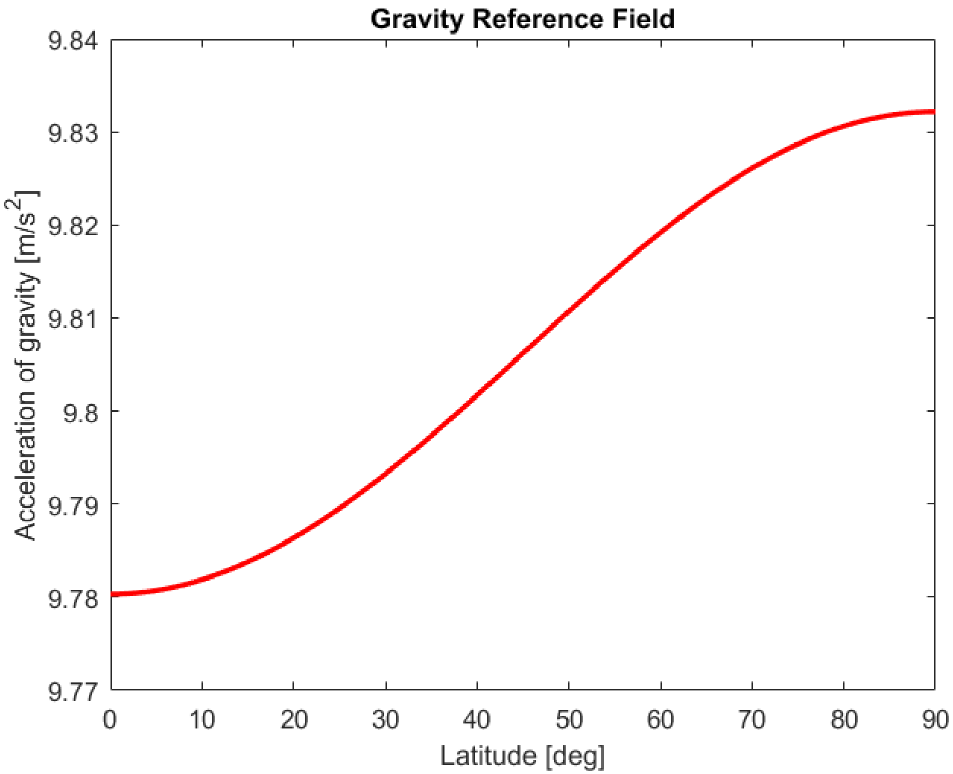
- Gravmag inversion is ill-posed and ambiguous
- Use information from other types of geophysical data as constraints

Gravity data

Measurement of the local acceleration of gravity

Factors that contributes to observed gravity:

- Mass of the earth
- Shape of the earth
- Rotation of the earth
- Elevation of the gravimeter above sea level
- Motion of the gravimeter
- Topography/terrain
- **Geology**



$$g_0 = 9.78031840(1 + 0.0053024 \sin^2 \theta + 0.0000058 \sin^2 2\theta) \text{ m/s}^2$$

Gravity corrections - get the geological effects

$$g_{geology} = g_{observed} - g_0 - g_{fa} - \underbrace{g_{slab} - g_{terr}}_{\text{Complete Bouguer}} - g_{iso} - g_{Eo}$$

$g_{geology} = \Delta g_z =$ gravity anomaly caused by **geology**

$g_{observed} =$ **observed** acceleration of gravity

$g_0 =$ earth ellipsoid

$g_{fa} = -0.3086 \times 10^{-2} h =$ free-air (gravimeter elevation) correction

$g_{slab} = 2\pi C_g \rho h =$ Bouguer slab correction

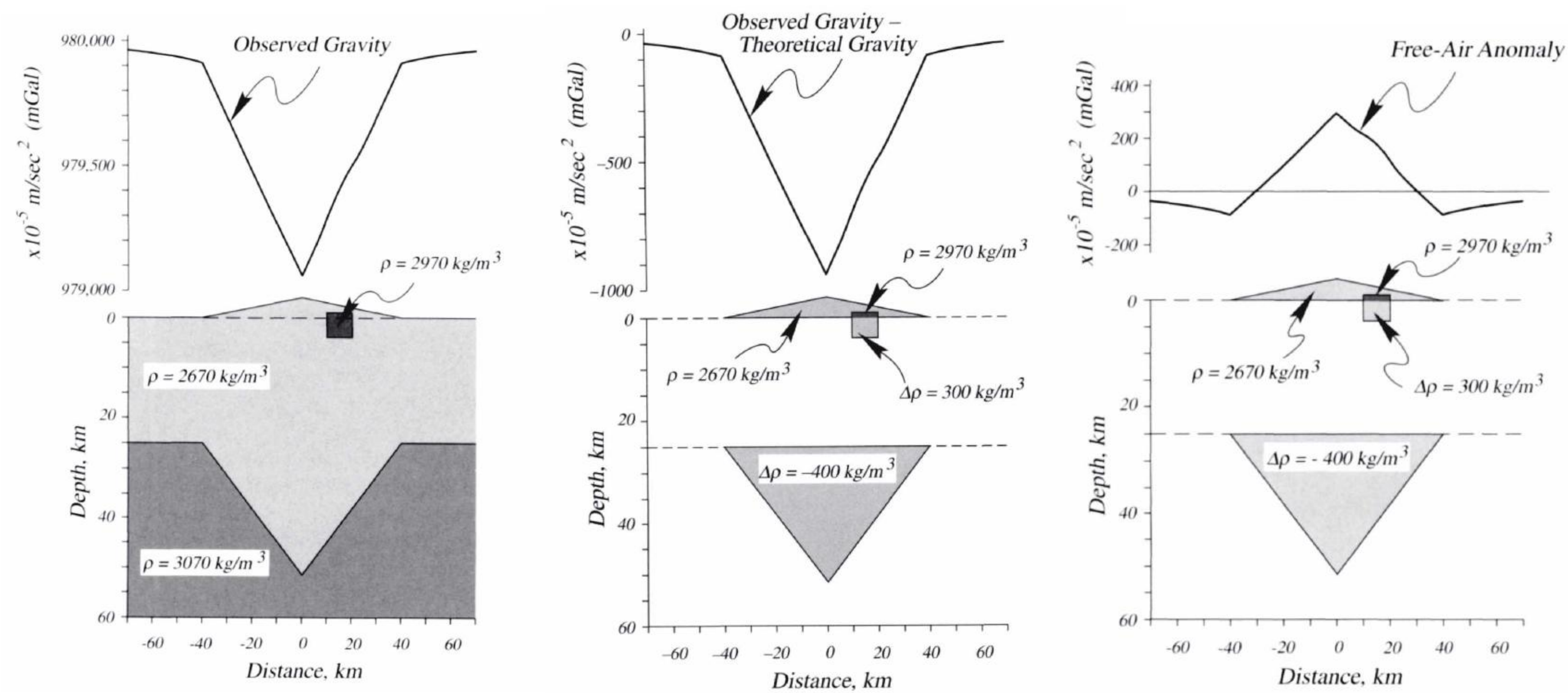
$g_{terr} =$ terrain corretion

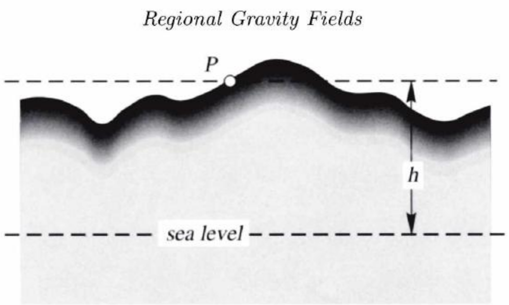
$g_{Eo} =$ Eotvos (moving gravimeter) correction

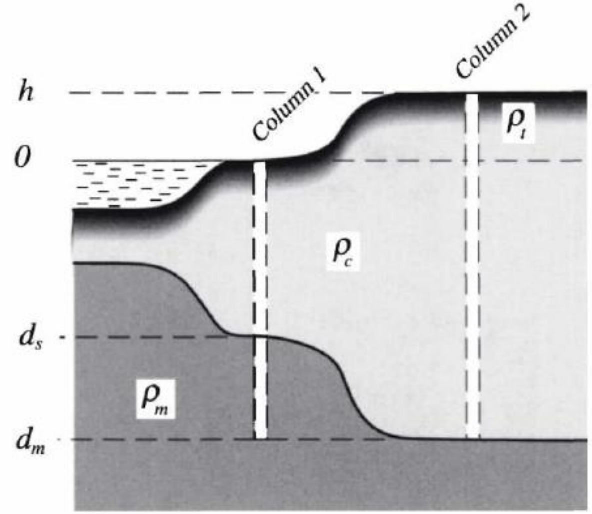
$g_{iso} =$ isostatic support correction

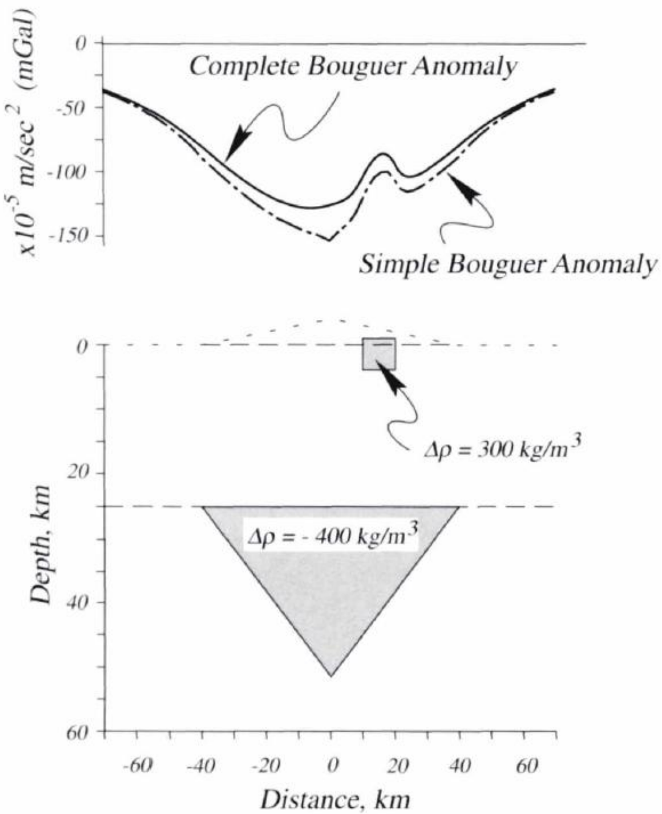
Gravity corrections

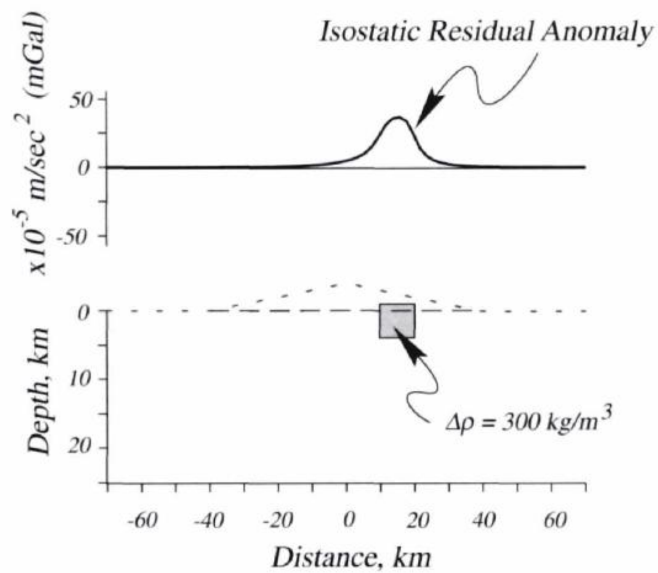
Figures from Blakely, 1996, Potential Theory in Gravity and Magnetic Applications



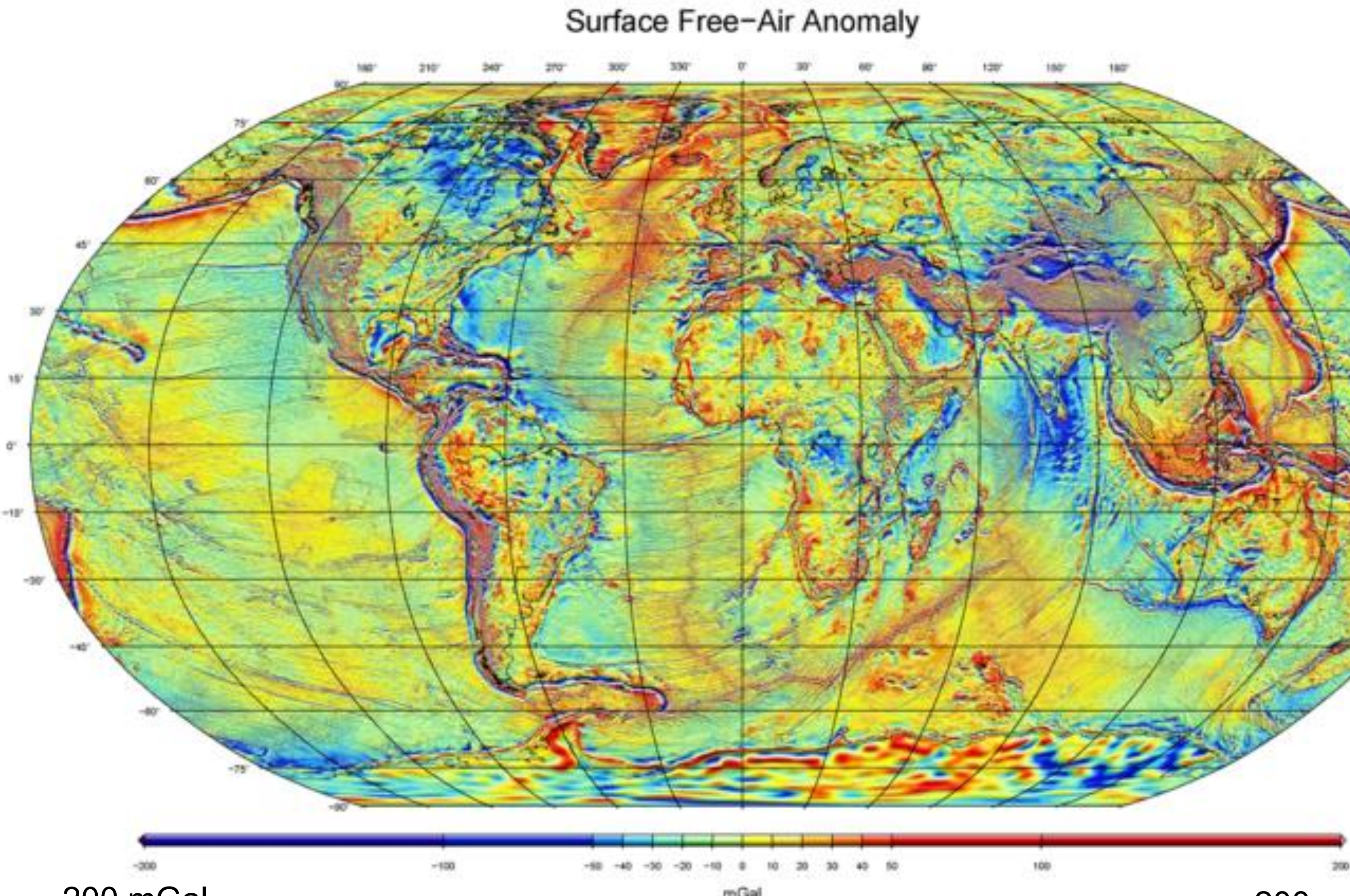








This is the anomaly Δg_z input to inversion

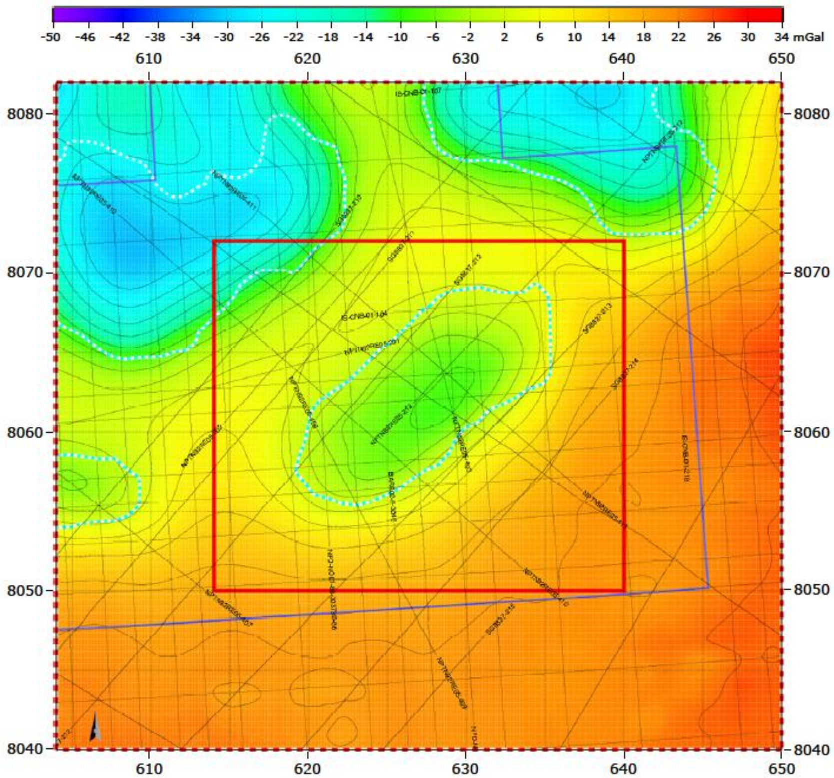


-200 mGal

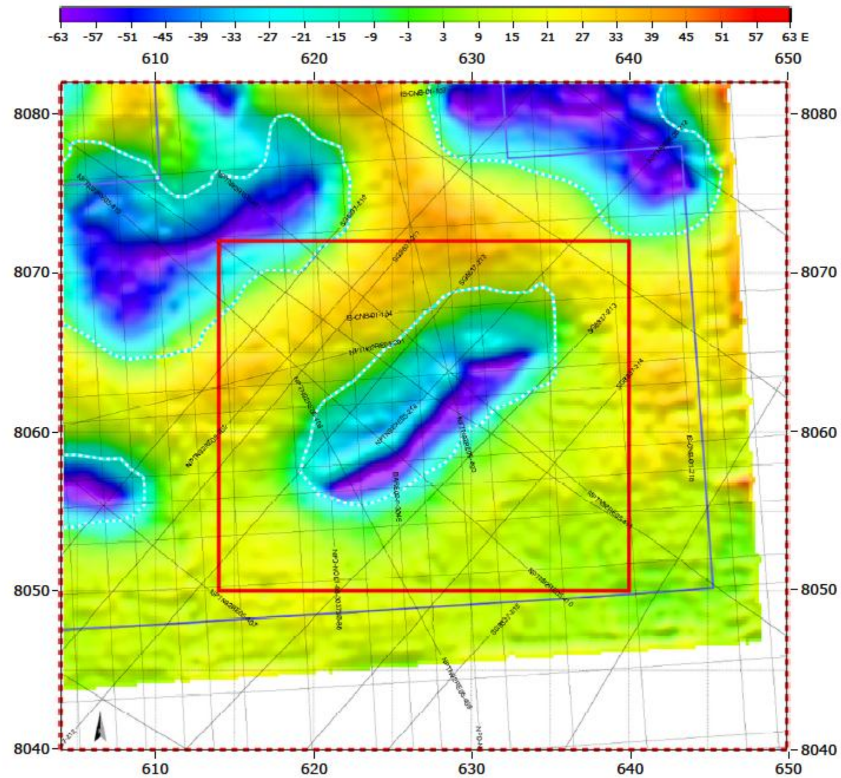
200 mGal

- Observed gravity: $g_z \approx 9.8\ \mathrm{m/s^2} = 980000\ \mathrm{mGal}$
- Geological effects: $g_{geology} \sim 0.001\ \mathrm{m/s^2} = 100\ \mathrm{mGal}$

Gravity and FTG data



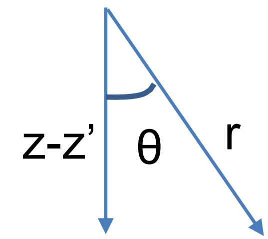
Free air Gravity (g_z)



Full Tensor Gravity ($g_{zz} = \partial_z g_z$)

Gravity data from a salt basin in the Barents Sea

3D gravity inversion (1)



Gravity inversion is a linear inverse problem (in density):

$$\mathbf{d} = \mathbf{A}\mathbf{m} + \mathbf{n}$$

$$m_j = \Delta\rho(\mathbf{x}_j)$$

$$\begin{aligned} d_i &= \Delta g_z(\mathbf{x}_i), \\ A_{ij} &= -C_g \frac{z_i - z_j}{|\mathbf{x}_i - \mathbf{x}_j|^3} \Delta V_j \end{aligned}$$

Newton’s law of gravity:

$$\Delta g_z(\boldsymbol{x}) = C_g \int \Delta\rho_z(\boldsymbol{x}') \frac{(z' - z)}{r^3}\, dV'$$

Objective function:

$$\Phi = \frac{1}{2}[\mathbf{d} - \mathbf{A}\mathbf{m}]^T \boldsymbol{\Sigma}_d^{-1} [\mathbf{d} - \mathbf{A}\mathbf{m}] + \frac{1}{2}\mathbf{m}^T \boldsymbol{\Sigma}_m^{-1} \mathbf{m}$$

3D gravity inversion (2)

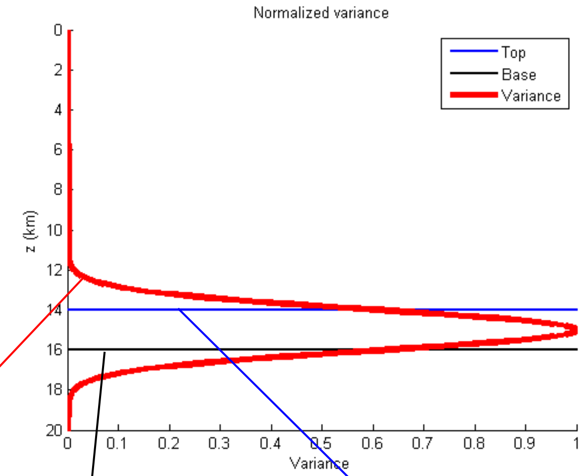
(Extended) Marquardt-Levenberg solution:

$$\mathbf{m} = [\mathbf{A}^T\mathbf{A} + \lambda\mathbf{J}]^{-1}\mathbf{A}^T\mathbf{d}$$

Constraints:

$$\begin{aligned}\mathbf{J} &= \mathbf{\Omega}^{-1}\mathrm{diag}(\mathbf{A}^T\mathbf{A}) \\ \lambda &= \frac{\sigma_d^2}{\sigma_m^2}.\end{aligned}$$

$$\Omega_j = \epsilon + (1-\epsilon)\mathrm{e}^{-(z_j-\tilde{z_j})^2/2\tilde{\sigma_j}^2}$$



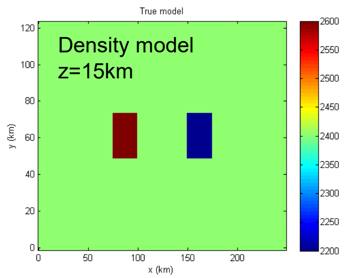
Base (Crust) horizon

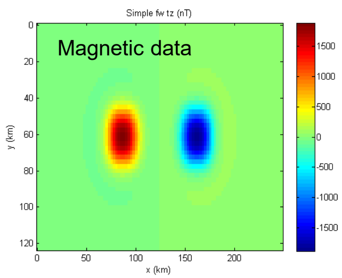
Top (Crust) horizon

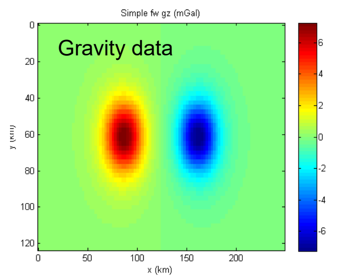
$$\begin{aligned}\tilde{z_j} &= \frac{1}{2}[Z_B(x_j,y_j) + Z_T(x_j,y_j)] \\ \tilde{\sigma_j} &= \frac{1}{2}[Z_B(x_j,y_j) - Z_T(x_j,y_j)]\end{aligned}$$

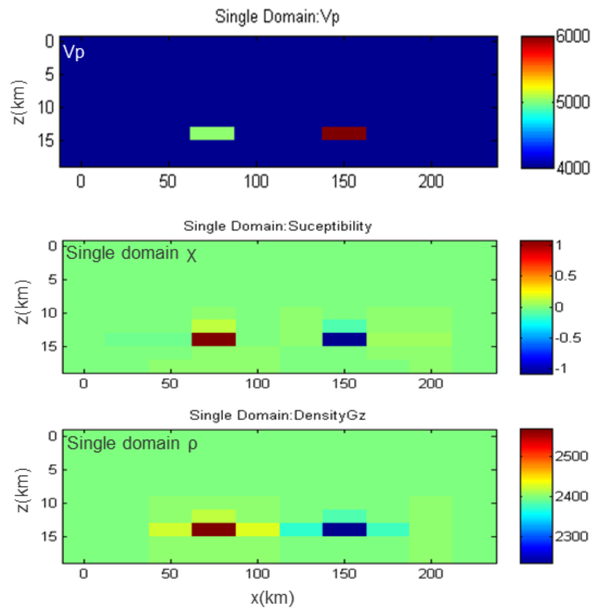
Inversion for density given geometry (horizons) is a linear inverse problem

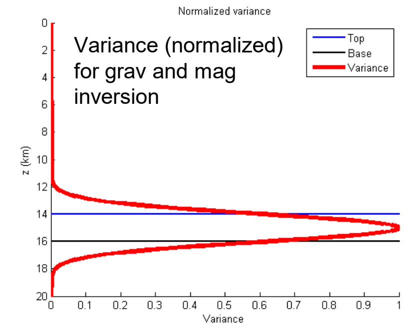
Synthetic example: GravMag inversion





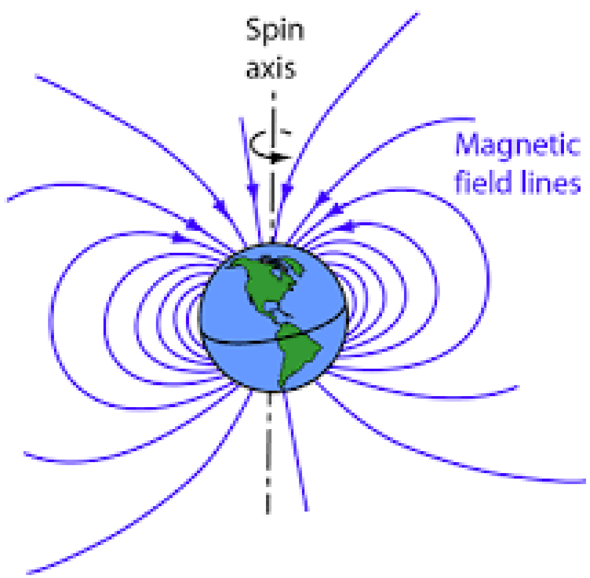






Seimic velocity model as structural constraint

Magnetic data



Usually we measure only the magnitude of the magnetic field vector $\boldsymbol{B}_{TMI}$

$$\boldsymbol{B}_{TMI} = |\boldsymbol{B}_0 + \boldsymbol{B}_A| = \sqrt{|\boldsymbol{B}_0|^2 + |\boldsymbol{B}_A|^2 + 2\boldsymbol{B}_0 \cdot \boldsymbol{B}_A}$$

Subtract the background field to get the anomaly

$$\boldsymbol{B}_A = |\boldsymbol{B}_0 + \boldsymbol{B}_A| - |\boldsymbol{B}_0|$$

$$B_A \simeq \boldsymbol{t} \cdot \boldsymbol{B}_A + \frac{|\boldsymbol{t} \times \boldsymbol{B}_A|^2}{|\boldsymbol{B}_0|}$$

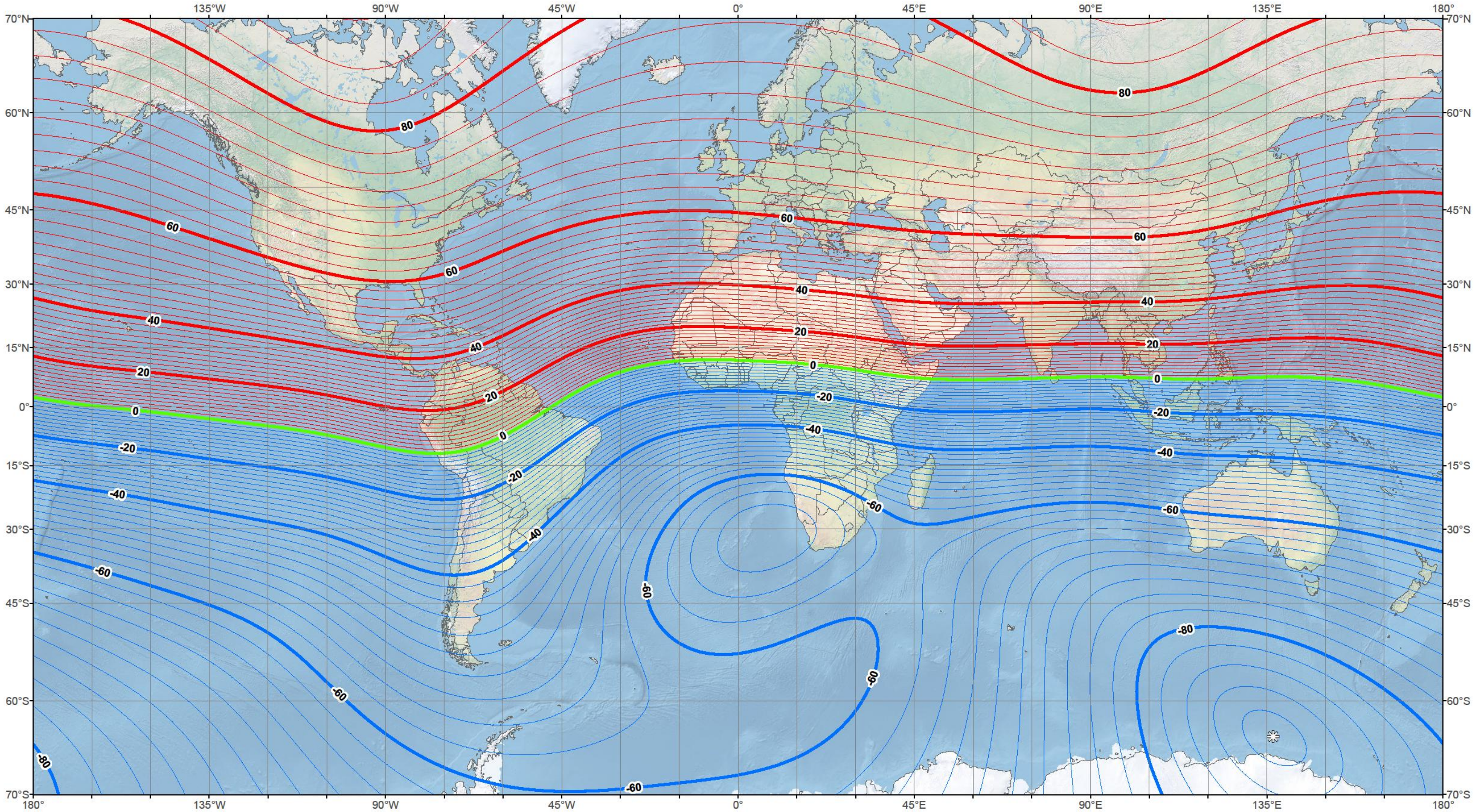
where $\boldsymbol{t}$ is the direction (tangent) of the background field

Magnetic Field X

Model Used:	IGRF2015
Latitude:	63.39532459° N
Longitude:	10.46789978° E
Elevation:	180.0 m GPS

Date	Declination (+ E | - W)	Inclination (+ D | - U)	Horizontal Intensity	North Comp (+ N | - S)	East Comp (+ E | - W)	Vertical Comp (+ D | - U)	Total Field
2019-08-22	3.6970°	74.8207°	13,598.0 nT	13,569.7 nT	876.8 nT	50,120.6 nT	51,932.4 nT
Change/year	0.2042°/yr	0.0114°/yr	-1.8 nT/yr	-4.9 nT/yr	48.3 nT/yr	33.1 nT/yr	31.5 nT/yr

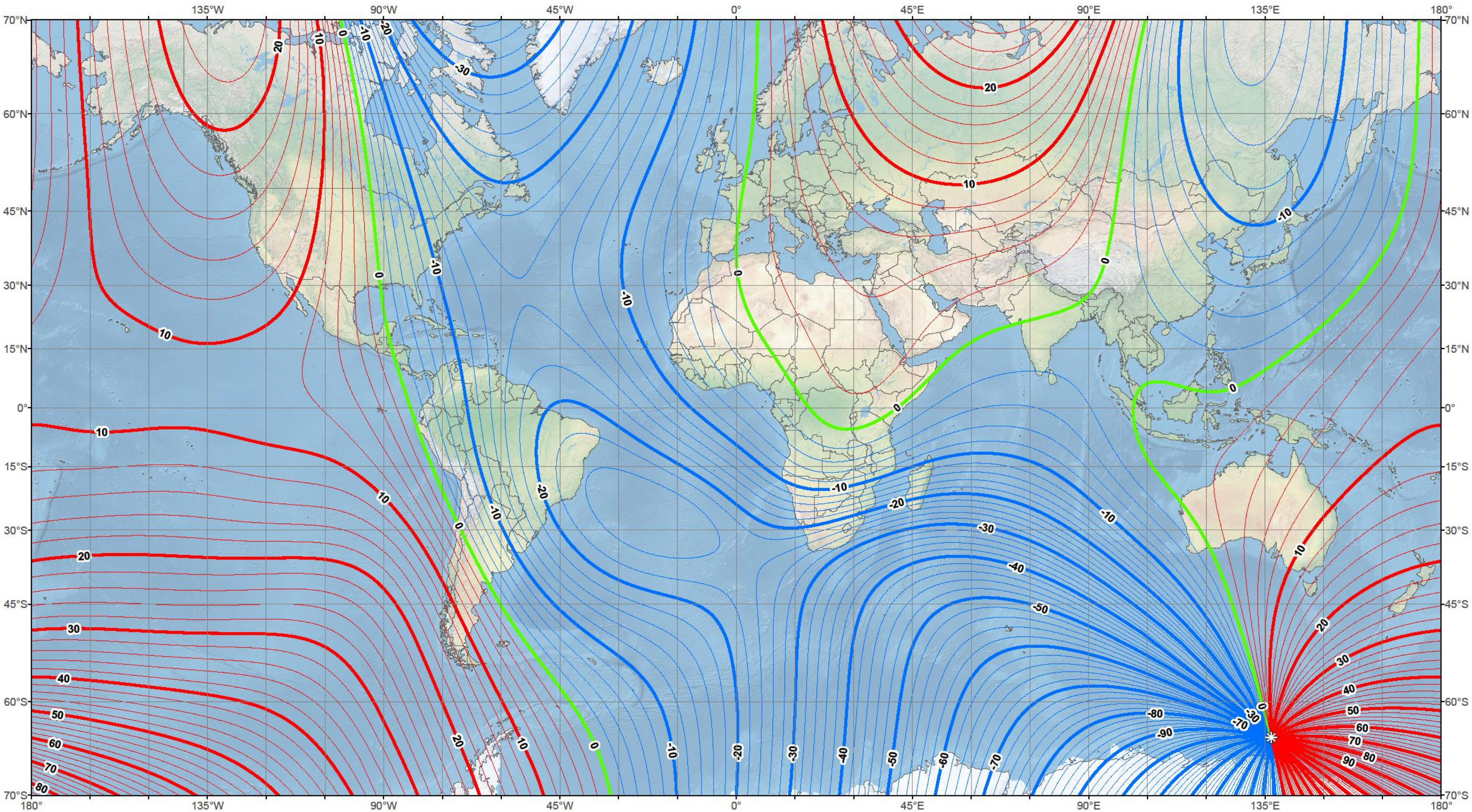
US/UK World Magnetic Model - Epoch 2015.0
Main Field Inclination (I)



Main field inclination (I)
Contour interval: 2 degrees, red contours positive (down); blue negative (up); green zero line.
Mercator Projection.
: Position of dip poles

Map developed by NOAA/NGDC & CIRES
http://ngdc.noaa.gov/geomag/WMM
Map reviewed by NGA and BGS
Published December 2014

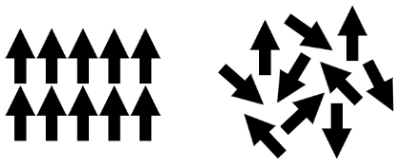
US/UK World Magnetic Model - Epoch 2015.0
Main Field Declination (D)

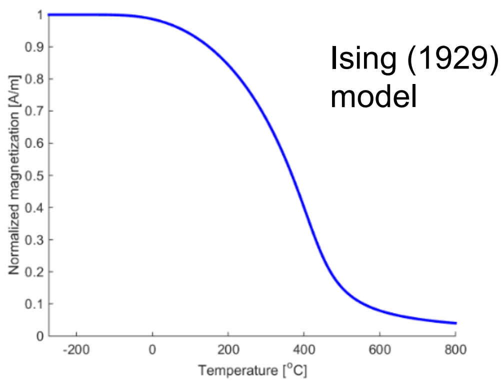


Main field declination (D)
Contour interval: 2 degrees, red contours positive (east); blue negative (west); green (agonic) zero line.
Mercator Projection.
: Position of dip poles

Map developed by NOAA/NGDC & CIRES
http://ngdc.noaa.gov/geomag/WMM
Map reviewed by NGA and BGS
Published December 2014

Magnetization

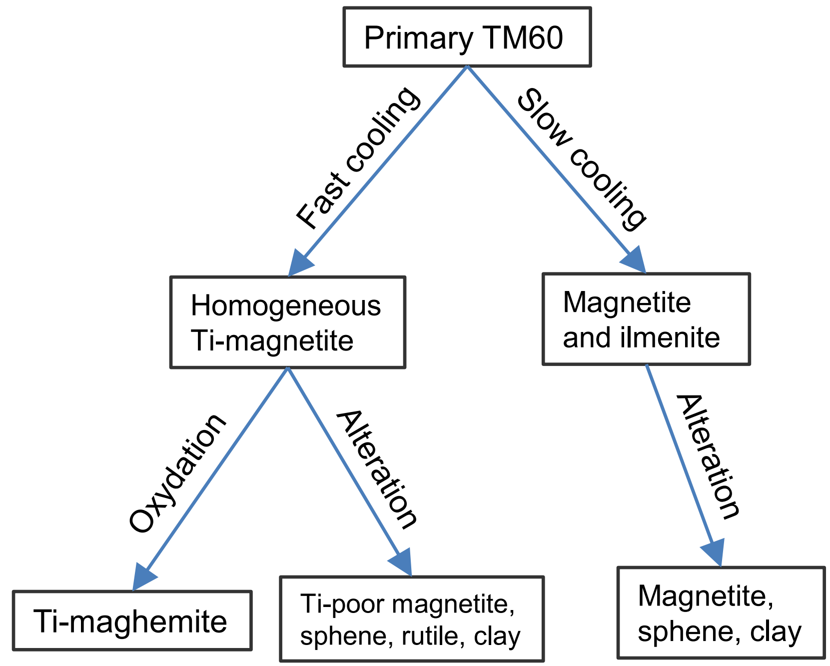




$$\mathcal{H} = -J\sum_{<ij>}\sigma_i\sigma_j - \mu_B H\sum_j \sigma_j$$

$$\frac{M}{M_0} = \tanh\left[\frac{T_C M}{T M_0} + \frac{CH}{T M_0}\right]$$

$$\boldsymbol{M} = \boldsymbol{M_R} + \boldsymbol{M_I} = \boldsymbol{M_R} + \chi\boldsymbol{H}$$



Adapted and simplified from Pariso and Johnson (1990)
Alteration in Costa Rica Ridge ODP data

From Ampere's law to potential-field magnetic

Amperes law

$$\nabla \times \mathbf{H} = \mathbf{j} + \epsilon \frac{\partial \mathbf{E}}{\partial t} = \mathbf{J}$$

If no macroscopic currents; $\mathbf{J} = 0$

$$\nabla \times \mathbf{H} = 0$$

Then $\mathbf{H}$ can be expressed as the gradient of a scalar potential ψ

$$\mathbf{H} = -\nabla \psi$$

Since (identity)

$$\nabla \times (\nabla \psi) = \mathbf{0}$$

Gauss law (no monopoles)

$$\nabla \cdot \mathbf{B} = 0$$

Magnetic flux (true magnetic field)

$$\mathbf{B} = \mu_0 (\mathbf{H} + \mathbf{M})$$

Poisson equation for ψ

$$\nabla^2 \psi = \nabla \cdot \mathbf{M}$$

Solution for the potential ψ

$$\psi(\mathbf{x}) = -\frac{1}{4\pi} \int \mathbf{M}(\mathbf{x}') \cdot \nabla' \frac{1}{|\mathbf{x}' - \mathbf{x}|} dV'$$

Solution for the magnetic field

$$\mathbf{B}(\mathbf{x}) = \mu_0 \mathbf{H}(\mathbf{x}) = -\mu_0 \nabla \psi(\mathbf{x})$$

Magnetic inversion for NRM (1)

Magnetic anomaly

$$\boldsymbol{B_A}(\boldsymbol{x}) = -\frac{\mu_0}{4\pi}\boldsymbol{\nabla\nabla}\cdot\iiint\frac{\boldsymbol{M}(\boldsymbol{x}')}{|\boldsymbol{x}'-\boldsymbol{x}|}dV'$$

$$\boldsymbol{M} = \boldsymbol{M_R} + \boldsymbol{M_I} = \boldsymbol{M_R} + \chi\boldsymbol{H}$$

Approximation: Assume $\boldsymbol{M}$ is independent of z or average over z

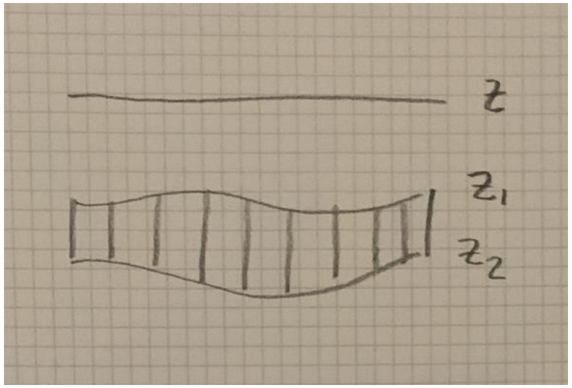
$$\boldsymbol{M}(\mathrm{x},\mathrm{y}) = \frac{1}{z_2-z_1}\int_{z_1}^{z_2}\boldsymbol{M}(x,y,z)\,dz$$

Then we can perform the integration over z in the equation for $\boldsymbol{B_A}$

$$\int_{z_1}^{z_2}\frac{1}{r}dz' = \ln[(z'-z)+r]\,|_{z_1}^{z_2} = F(z_2)-F(z_1)$$

$$F(z') = \ln[(z'-z)+r]$$

$$r = |\boldsymbol{x}'-\boldsymbol{x}|$$



Magnetic inversion for NRM (2)

Forward model

$$B_A(\boldsymbol{x}) \simeq \boldsymbol{t} \cdot \boldsymbol{B_A}(\boldsymbol{x}) = \frac{\mu_0}{4\pi} \sum_{\boldsymbol{i=1}}^{\boldsymbol{3}} \sum_{\boldsymbol{j=1}}^{\boldsymbol{3}} \iint t_i A_{ij}(\boldsymbol{x}, \boldsymbol{x}') M_j(x', y')\, dx' dy'$$

$$A_{ij} = \frac{\partial^2 F(z_2)}{\partial x_i \partial x_j} - \frac{\partial^2 F(z_1)}{\partial x_i \partial x_j}$$

Discretize and prepare for inversion

$$B_A(\boldsymbol{x}_i) = \sum_j Q_M\left(\boldsymbol{x}_i, \boldsymbol{x}_j'\right) M\left(\boldsymbol{x}_j'\right)$$

$$\boldsymbol{M} = M\boldsymbol{e}$$

$$M_j = Me_j$$

$$Q_M\left(\boldsymbol{x}_i, \boldsymbol{x}_j'\right) = \frac{\mu_0}{4\pi} \Delta S_j' \sum_{k=1}^{3} \sum_{l=1}^{3} t_k A_{kl}\left(\boldsymbol{x}_i, \boldsymbol{x}_j'\right) e_{l,}$$

$$\boldsymbol{A_{ij} = \frac{\partial^2 F(z_2)}{\partial x_i \partial x_j} - \frac{\partial^2 F(z_1)}{\partial x_i \partial x_j}}$$

$$\boldsymbol{\frac{\partial^2 F(z')}{\partial x_i \partial x_j} = -\frac{1}{[(z'-z)+r]^2}\left[\delta_{i3} + \frac{(x_i'-x_i)}{r}\right]\left[\delta_{j3} + \frac{(x_j'-x_j)}{r}\right] + \frac{1}{[(z'-z)+r]}\left[\frac{\delta_{ij}}{r} - \frac{(x_i'-x_i)(x_j'-x_j)}{r^3}\right]}$$

Magnetic inversion for NRM (3)

Linear inversion; standard form

$$\mathbf{d} = \mathbf{Lm} + \mathbf{n}$$
$$\Phi = \frac{1}{2}[\mathbf{d} - \mathbf{Lm}]^{\mathbf{T}}[\mathbf{d} - \mathbf{Lm}] + \lambda\frac{1}{2}\mathbf{m}^{\mathbf{T}}\mathbf{D}^{\mathbf{T}}\mathbf{Dm}$$

$$d_i = B_A(\boldsymbol{x}_i)$$
$$\mathrm{m}_j = M(\boldsymbol{x}'_j)$$
$$L_{ij} = Q(\boldsymbol{x}_i, \boldsymbol{x}'_j)$$

Marquardt-Levenberg solution

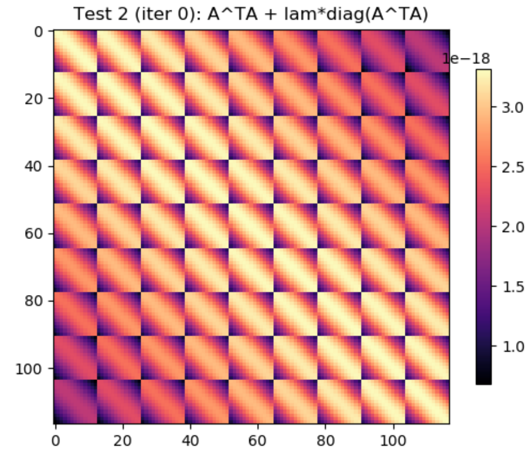
$$\mathbf{m} = [\mathbf{L}^{\mathbf{T}}\mathbf{L} + \lambda\mathbf{D}^{\mathbf{T}}\mathbf{D}]^{-1}\mathbf{L}^{\mathbf{T}}\mathbf{d}$$

$$\mathbf{D}^{\mathbf{T}}\mathbf{D} = \mathrm{diag}(\mathbf{L}^{\mathbf{T}}\mathbf{L})$$

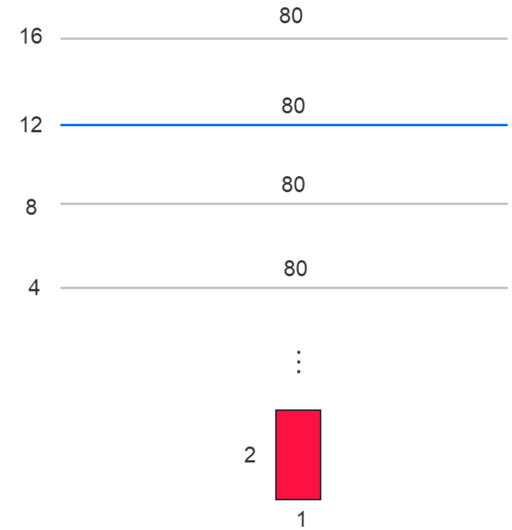
Simple example

- Grid spacing: 250m
- Recording altitude: 3000m

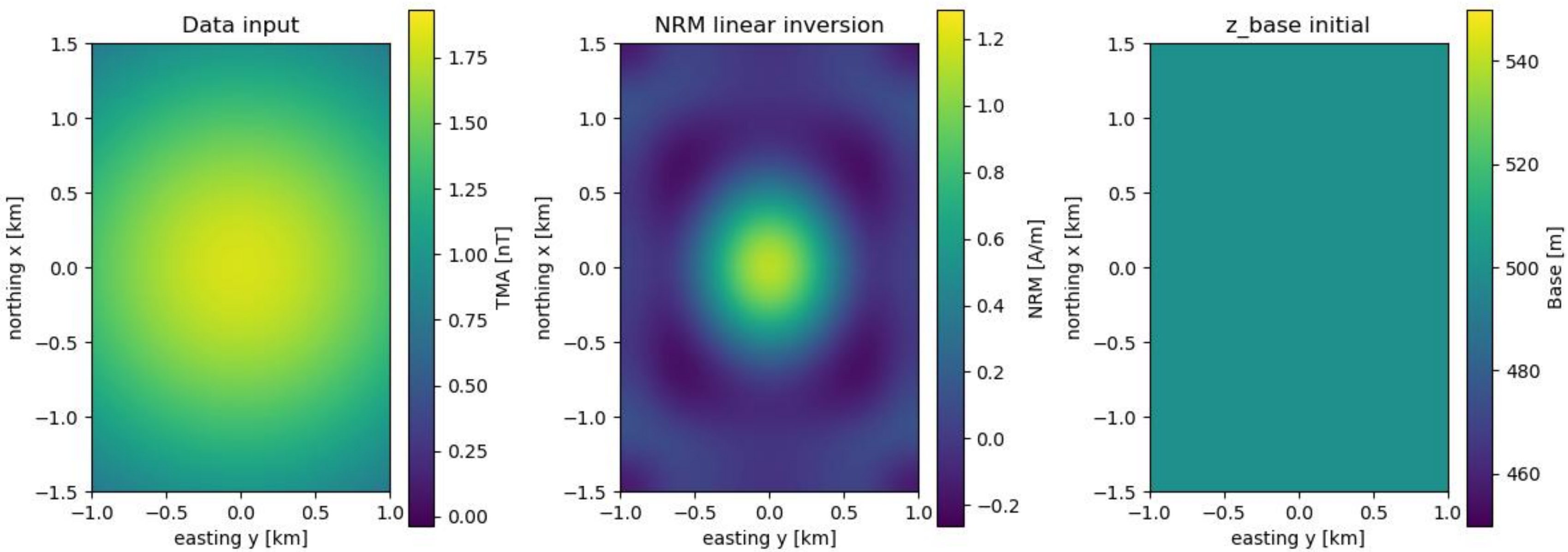
- Try it yourself (Python code)



Model and recording surface in number of grid cells



Magnetic test 2:



For an $n \times n$ matrix A

$$\text{rank}(A) + \ker(A) = n$$

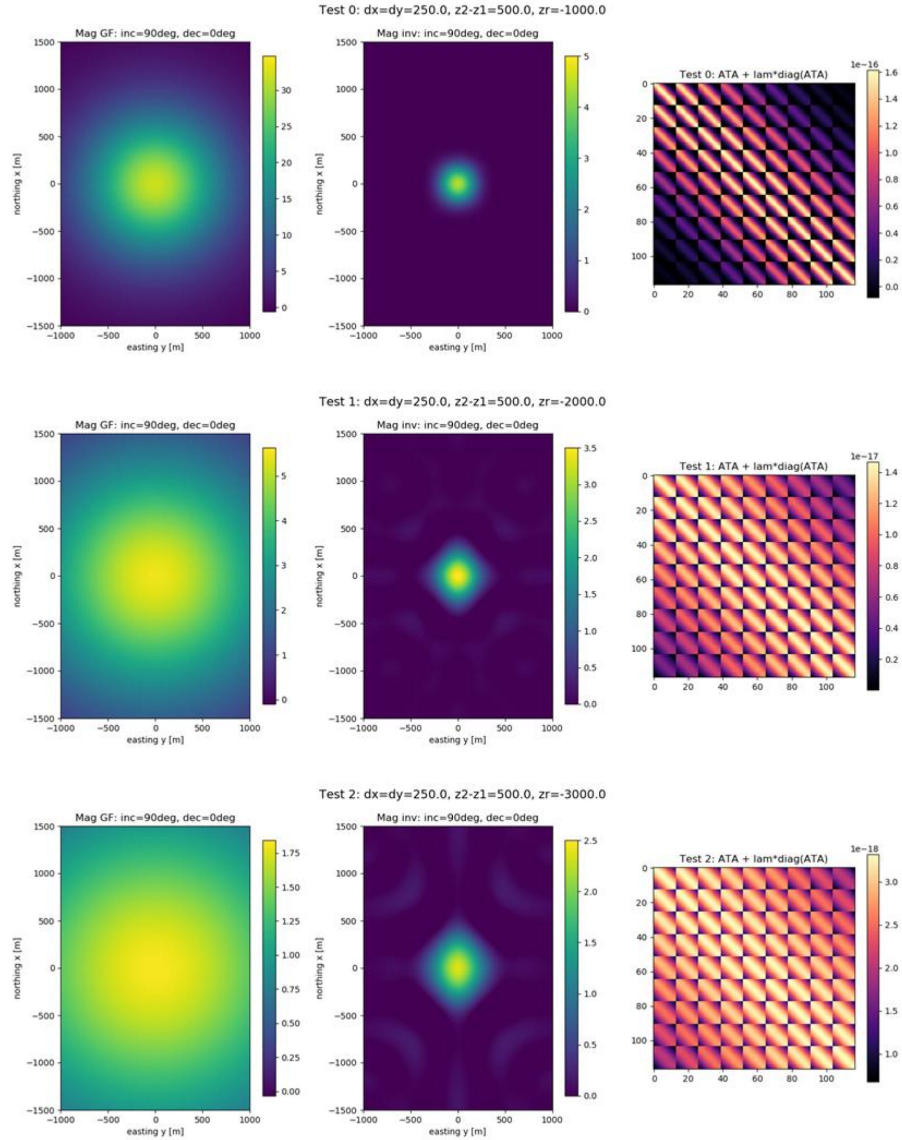
rank(A) = number of linearly independent rows
= dimension of vector-space spanned by rows/columns of A

Kernel: $A x_0 = 0$
(null space) $x_0 \in \ker(A)$

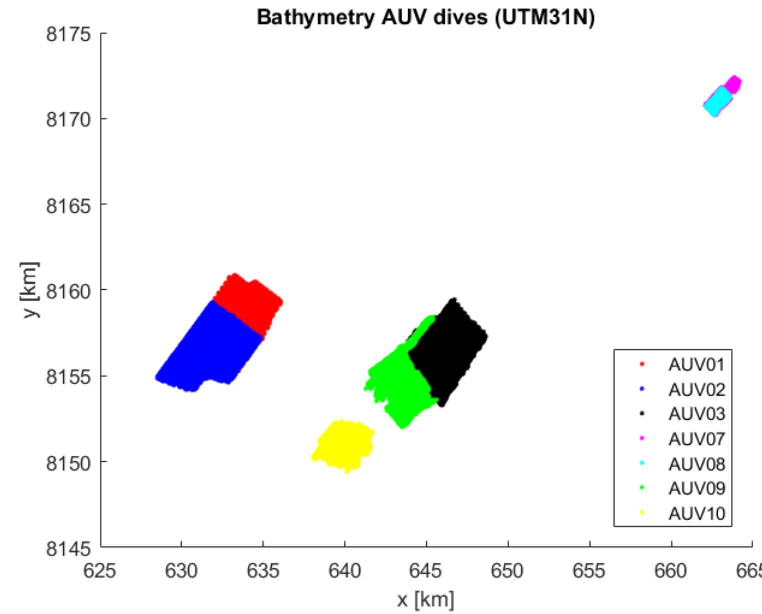
Then

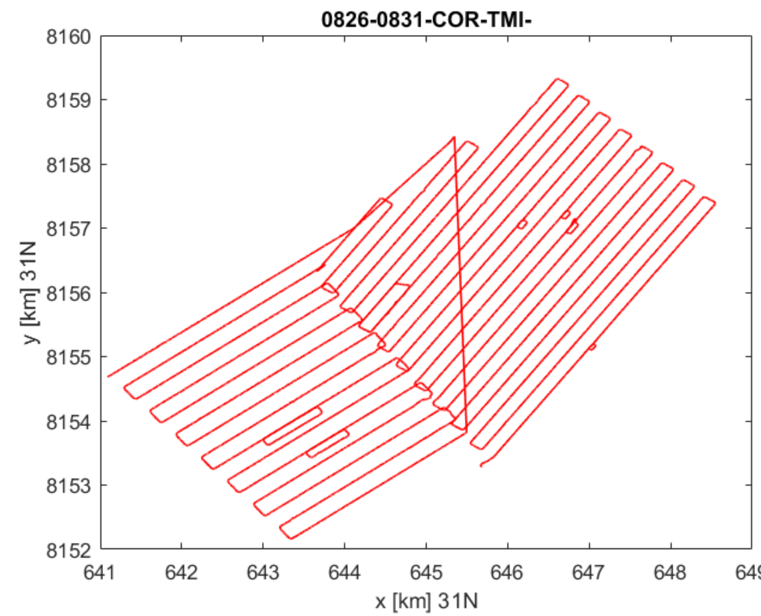
$$A(x + \alpha x_0) = Ax = b$$

because $A x_0 = 0$



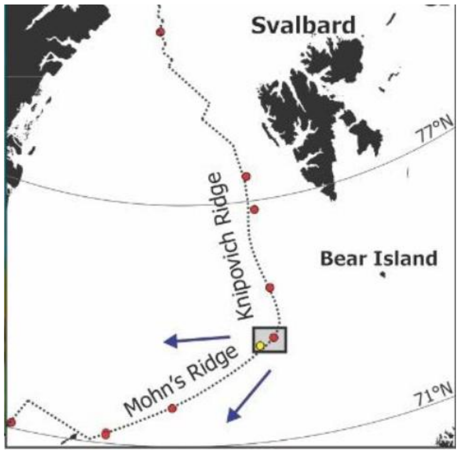
Hugin AUV

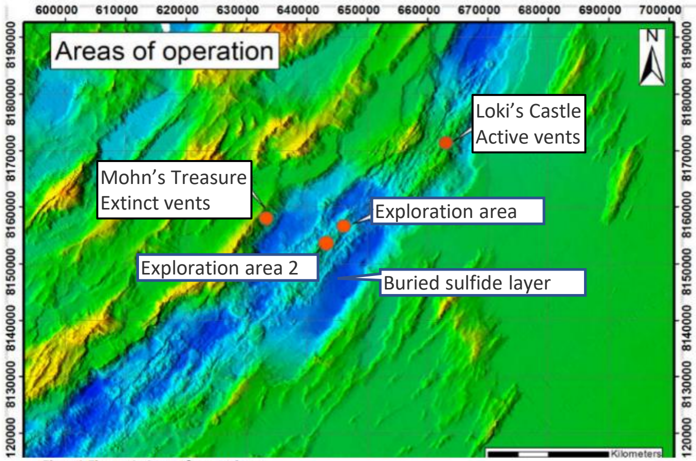




AUV track explo-area 1

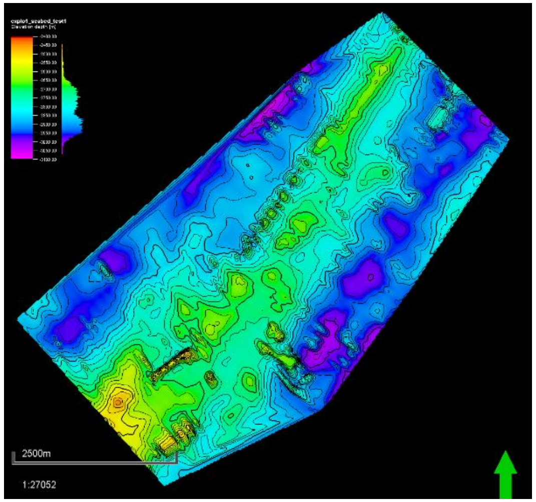
AUV-magnetic example



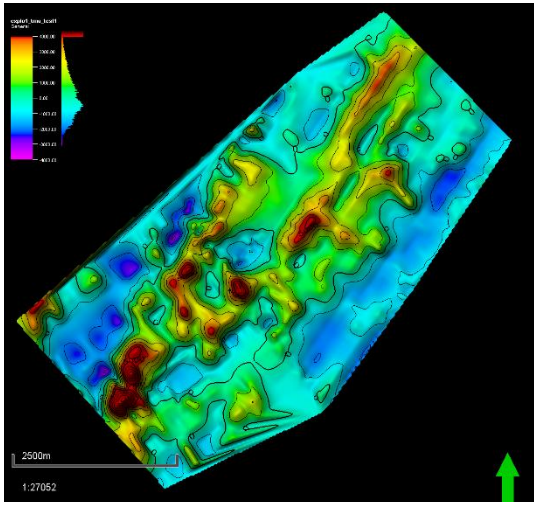


Figur 1 Figure 1. Area of operations.

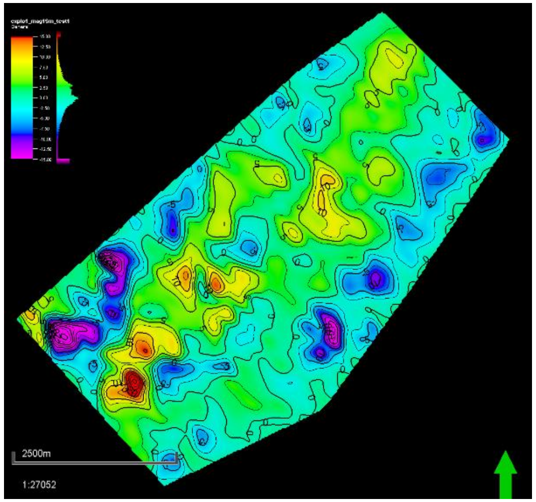
Data from NTNU MarMine cruise to Mohn's Ridge 2016 (Lim et al., 2019; Lim PhD thesis, 2020)



Bathymetry

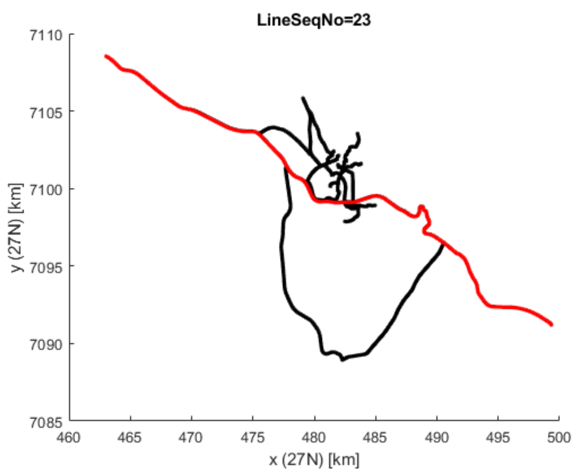


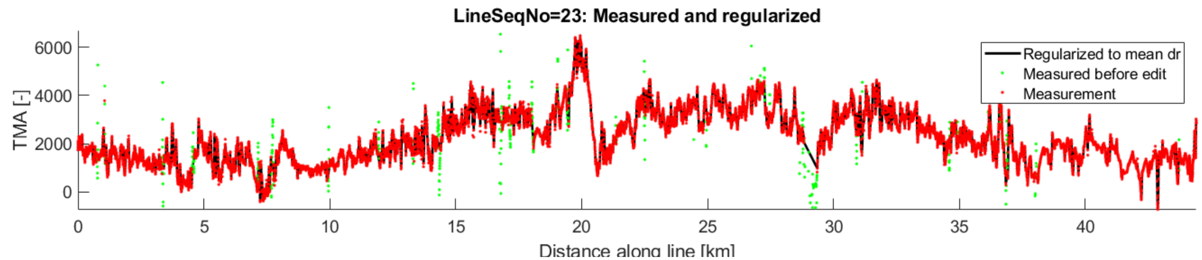
Total magnetic anomaly (TMA)



Magnetization from inversion

Iceland high-res magnetics





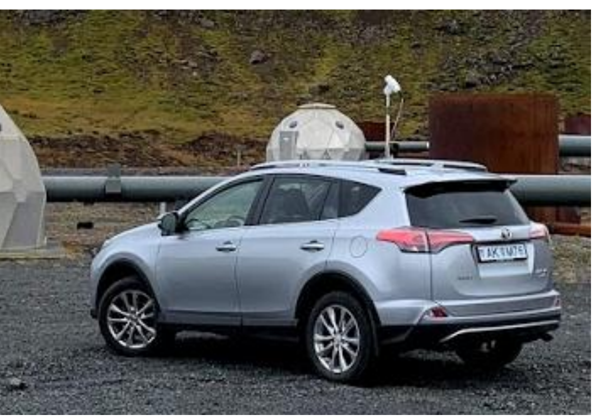
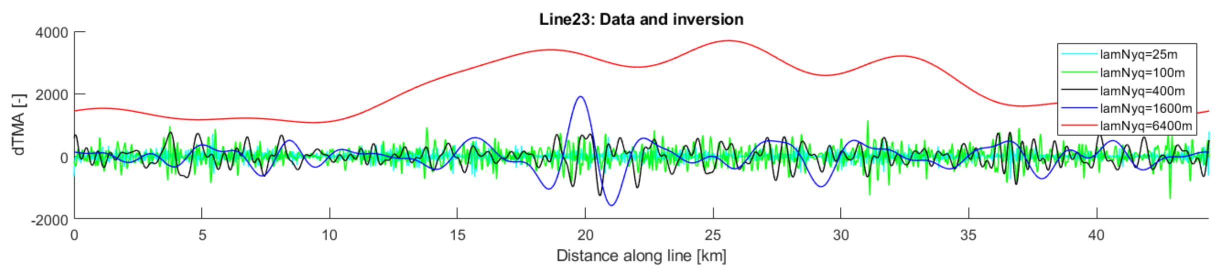
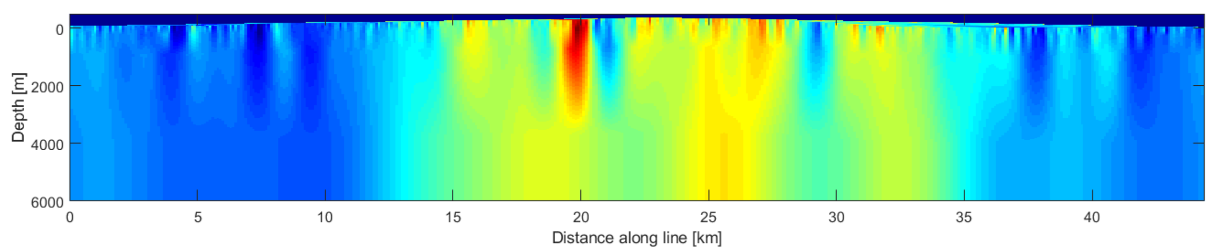
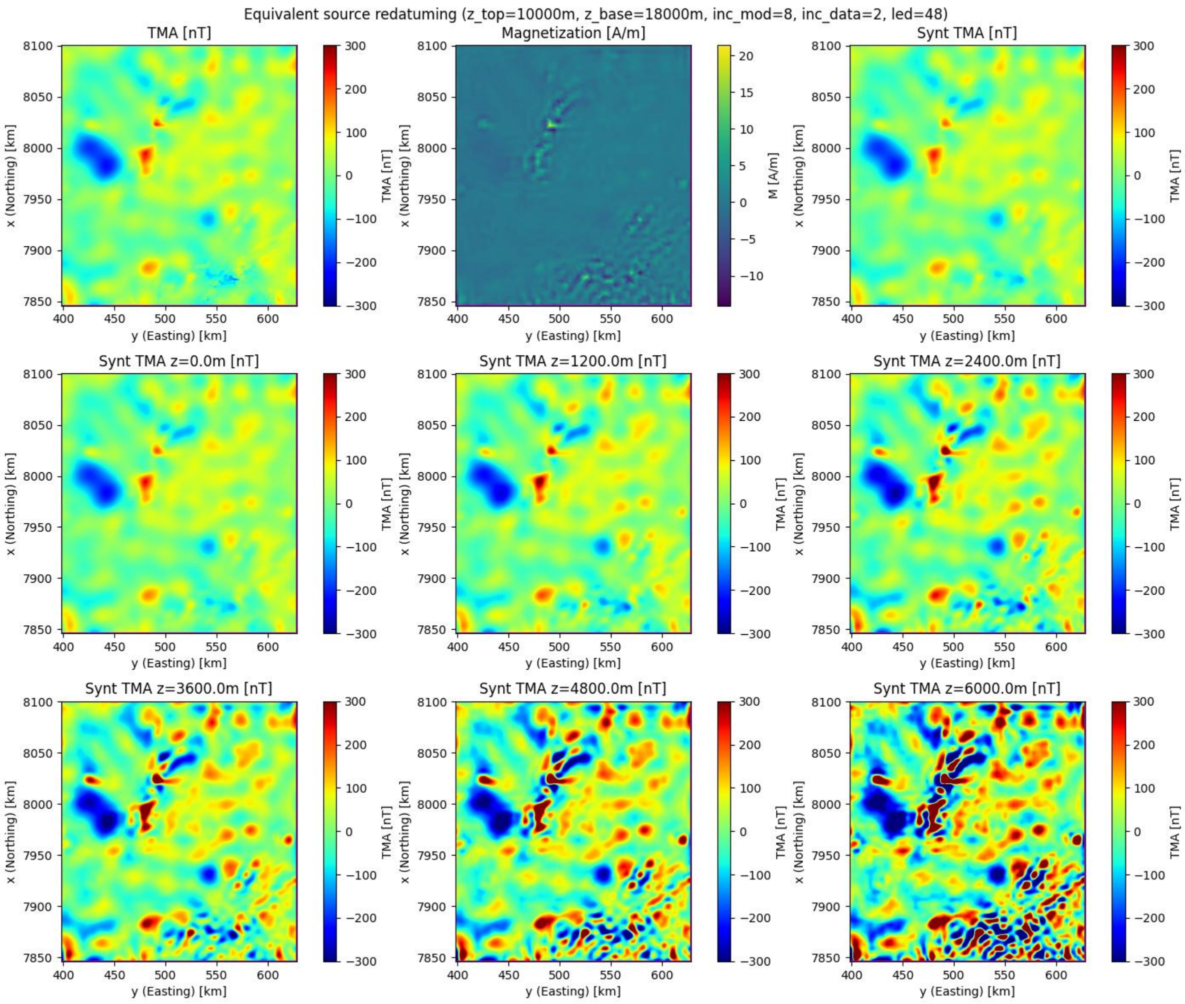


Photo: Bjørn M. Sæther

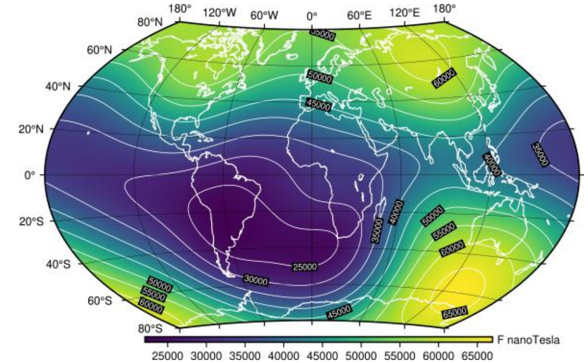




In-field referencing: Equivalent-source method



- Drilling of horizontal wells
- Geosteering with magnetic measurements
- Global reference model
  - Magnitude
  - Inclination
  - Declination
- Local anomalies



Posterior expectation and variance

- Compute the «full» pdf
- Non-linear (non-Gaussian): Monte Carlo simulation (MCMC)
- Linear (Gaussian): Analytical formulas can be used (Buland and Omre)

Posterior expectation

$$\mu_{\mathbf{m}|\mathbf{d}} = E(\mathbf{m}) = \int \mathbf{m} f(\mathbf{m}|\mathbf{d}) d\mathbf{m}$$

Posterior covariance

$$\mathbf{\Sigma} = E[(\mathbf{m} - \mu_{\mathbf{m}|\mathbf{d}})(\mathbf{m} - \mu_{\mathbf{m}|\mathbf{d}})^T]$$

- MAP solution: Maximum of $f(\mathbf{m}|\mathbf{d})$

$$\hat{\mathbf{m}}_{MAP} = \underset{\mathbf{m}}{\operatorname{argmax}} f(\mathbf{m}|\mathbf{d})$$

Bayesian AVO inversion (Buland and Omre)

Aki and Richards approximation to Zoeppritz equations:

$$c_{PP} = a_\alpha \frac{\Delta\alpha}{\alpha} + a_\beta \frac{\Delta\beta}{\beta} + a_\rho \frac{\Delta\rho}{\rho}$$

$$\begin{aligned}
a_\alpha(t,\theta) &= \frac{1}{2}(1+\tan\theta) \\
a_\beta(t,\theta) &= -4\left(\frac{\bar{\beta}}{\bar{\alpha}}\right)^2 \sin^2\theta \\
a_\rho(t,\theta) &= \frac{1}{2}\left(1-4\left(\frac{\bar{\beta}}{\bar{\alpha}}\right)^2 \sin^2\theta\right)
\end{aligned}$$

Linear forward model $\Rightarrow$ linear inverse problem

Bayesian AVO inversion (Buland and Omre)

Aki and Richards approximation to Zoeppritz equations:

$$c_{PP} = a_\alpha \frac{\Delta\alpha}{\alpha} + a_\beta \frac{\Delta\beta}{\beta} + a_\rho \frac{\Delta\rho}{\rho}$$

$$\begin{aligned} \alpha &= \text{P-wave velocity} \\ \beta &= \text{S-wave velocity} \\ \rho &= \text{mass density} \end{aligned}$$

$$\begin{aligned} a_\alpha(t,\theta) &= \frac{1}{2}(1+\tan\theta) \\ a_\beta(t,\theta) &= -4\left(\frac{\bar{\beta}}{\bar{\alpha}}\right)^2 \sin^2\theta \\ a_\rho(t,\theta) &= \frac{1}{2}\left(1-4\left(\frac{\bar{\beta}}{\bar{\alpha}}\right)^2 \sin^2\theta\right) \end{aligned}$$

$$\alpha = \frac{1}{2}(v_{P2} + v_{P1})$$

$$\Delta\alpha = \frac{1}{2}(v_{P2} - v_{P1})$$

Linear forward model $\Rightarrow$ linear inverse problem

Seismic two-way time domain:

$$\frac{\partial}{\partial t}\ln\alpha \quad = \quad \frac{1}{\alpha}\frac{\partial\alpha}{\partial t} \simeq \frac{1}{\Delta t}\frac{\Delta\alpha}{\alpha}$$

and similar for β and ρ

Hence, rewrite the Aki and Richards approximation as

$$c_{PP} \quad = \quad a_\alpha \frac{\partial}{\partial t}\ln\alpha + a_\beta \frac{\partial}{\partial t}\ln\beta + a_\rho \frac{\partial}{\partial t}\ln\rho$$

Matrix-vector form:

$$\begin{aligned}\mathbf{c} &= \mathbf{A}\mathbf{m}' \\ \mathbf{m}' &= \frac{\partial \mathbf{m}}{\partial t}\end{aligned}$$

Model parameters:

$$\begin{aligned}\mathbf{m}(t) &= [\ln\alpha(t), \ln\beta(t), \ln\rho(t)]^T \\ E[\mathbf{m}(t)] &= \mu(t) = [\mu_\alpha(t), \mu_\beta(t), \mu_\rho(t)]^T\end{aligned}$$

Covariance:

$$\begin{aligned}\mathbf{\Sigma}(t,s) &= \text{cov}(\mathbf{m}(t),\mathbf{m}(s)) \\ \mathbf{\Sigma}(t,s) &= \mathbf{\Sigma}_0\nu(\tau), \quad \tau = t-s\end{aligned}$$

Parameter covariance (symmetric):

$$\mathbf{\Sigma}_0 = \begin{bmatrix} \sigma_\alpha^2 & \sigma_\alpha\sigma_\beta\nu_{\alpha\beta} & \sigma_\alpha\sigma_\rho\nu_{\alpha\rho} \\ & \sigma_\beta^2 & \sigma_\beta\sigma_\rho\nu_{\beta\rho} \\ & & \sigma_\rho^2 \end{bmatrix}$$

Spatial (in seismic time) correlation:

$$\begin{aligned}\nu(\tau) &= e^{-(\tau/r)^2} \\ r &= \text{correlation length}\end{aligned}$$

Prior expectation, using the linear-operator property:

$$
\begin{aligned}
\partial_t \mathbf{m} &= \mathbf{m}' = [\partial_t \ln \alpha, \partial_t \ln \beta, \partial_t \ln \rho]^T \\
E[\partial_t \mathbf{m}] &= \partial_t E[\mathbf{m}] = \partial_t \mu(t)
\end{aligned}
$$

$$
\boxed{
\begin{aligned}
E[ax] &= aE[x] \\
E[a^2(x-\mu)^2] &= a^2E[(x-\mu)^2]
\end{aligned}
}
$$

Prior covariance:

$$
\begin{aligned}
\text{cov}(\mathbf{m}'(t), \mathbf{m}'(s)) &= \partial_t \partial_s \mathbf{\Sigma}(t,s) \\
&= -\mathbf{\Sigma}_0 \partial_\tau^2 \nu(\tau) = \mathbf{\Sigma}''
\end{aligned}
$$

Prior (*apriori*) distribution for $\mathbf{m}$:

$$
\begin{aligned}
\mathbf{m} &\sim \mathcal{N}(\mu, \mathbf{\Sigma}) \\
\partial_t \mathbf{m} &\sim \mathcal{N}(\mu', \mathbf{\Sigma}'')
\end{aligned}
$$

The distribution of $[\alpha, \beta, \rho]$ is log-normal.

Forward model (Likelihood; data misfit)

$$
\begin{aligned}
\mathbf{d} &= \mathbf{Sc} + \mathbf{n} = \mathbf{SAm}' + \mathbf{n} \\
\mathbf{S} &= \text{directional source} \\
\mathbf{d} &= \text{data after PSTM, CIP gathers or angle stacks} \\
\mathbf{n} &= \text{noise; error}
\end{aligned}
$$

Distributions:

$$
\begin{aligned}
\mathbf{n} &= \mathbf{d} - \mathbf{SAm}' \sim \mathcal{N}(0, \mathbf{\Sigma}_e) \\
\mathbf{d} &\sim \mathcal{N}(\mu_d, \mathbf{\Sigma}_d) \\
\mu_d &= \mathbf{SA}\mu_m' \\
\mathbf{\Sigma}_d &= (\mathbf{SA})\mathbf{\Sigma}_m''(\mathbf{SA})^T + \mathbf{\Sigma}_e
\end{aligned}
$$

$$
\boxed{
\begin{aligned}
E[ax] &= aE[x] \\
E[a^2(x-\mu)^2] &= a^2E[(x-\mu)^2]
\end{aligned}
}
$$

Joint distribution for $\mathbf{d}$ and $\mathbf{m}$:

$$\begin{bmatrix} \mathbf{m} \\ \mathbf{d} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_m \\ \mu_d \end{bmatrix}, \begin{bmatrix} \mathbf{\Sigma}_m & \mathbf{\Sigma}_{m,d} \\ \mathbf{\Sigma}_{m,d} & \mathbf{\Sigma}_d \end{bmatrix}\right)$$

Off-diagonal covariance:

$$\mathbf{\Sigma}_{d,m} = \mathsf{cov}(\mathbf{d}, \mathbf{m}) = \mathsf{cov}(\mathbf{SAm}', \mathbf{m}) = \mathbf{SA\Sigma}'_m$$

Posterior distribution:

$$\mathbf{m}|\mathbf{d} \sim \mathcal{N}(\mu_{\mathbf{m}|\mathbf{d}}, \mathbf{\Sigma}_{\mathbf{m}|\mathbf{d}})$$

Bayes theorem:

$$\begin{aligned} f(\mathbf{m}, \mathbf{d}) &= f(\mathbf{d}|\mathbf{m})f(\mathbf{m}) = f(\mathbf{m}|\mathbf{d})f(\mathbf{d}) \\ f(\mathbf{m}|\mathbf{d}) &\propto f(\mathbf{d}|\mathbf{m})f(\mathbf{m}) \end{aligned}$$

Posterior expectation:

$$\mu_{\mathbf{m}|\mathbf{d}} = \int \mathbf{m}\, f(\mathbf{m}|\mathbf{d}) d\mathbf{m}$$

Analytical solution for Gaussian (normal) distributions:

$$\begin{aligned} \mu_{\mathbf{m}|\mathbf{d}} &= \mu_m + \boldsymbol{\Sigma}_{\mathbf{m},\mathbf{d}} \boldsymbol{\Sigma}_d^{-1} (\mathbf{d}_{obs} - \mathbf{d}_{mod}) \\ \boldsymbol{\Sigma}_{\mathbf{m}|\mathbf{d}} &= \boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}_{m,d} \boldsymbol{\Sigma}_d^{-1} \boldsymbol{\Sigma}_{m,d}^T \end{aligned}$$

Applicable to linear inverse problems

Important observations:

$\mu_{\mathbf{m}|\mathbf{d}}$: update driven by data misfit

$\boldsymbol{\Sigma}_{\mathbf{m}|\mathbf{d}}$: independent of data

Substitute

$$\mathbf{\Sigma}_{m,d} = \mathbf{SA\Sigma}'_m$$

Then

$$\begin{aligned} \mu_{\mathbf{m}|\mathbf{d}} &= \mu_m + (\mathbf{SA\Sigma}'_m)\mathbf{\Sigma}_d^{-1}(\mathbf{d}_{obs} - \mathbf{d}_{mod}) \\ \mathbf{\Sigma}_{\mathbf{m}|\mathbf{d}} &= \mathbf{\Sigma}_m - (\mathbf{SA\Sigma}'_m)\mathbf{\Sigma}_d^{-1}(\mathbf{SA\Sigma}'_m)^T \end{aligned}$$

Data quality; noise:

$$\mathbf{\Sigma}_d \text{ is small} \quad \Rightarrow \quad \mathbf{\Sigma}_{\mathbf{m}|\mathbf{d}} \ll \mathbf{\Sigma}_m$$

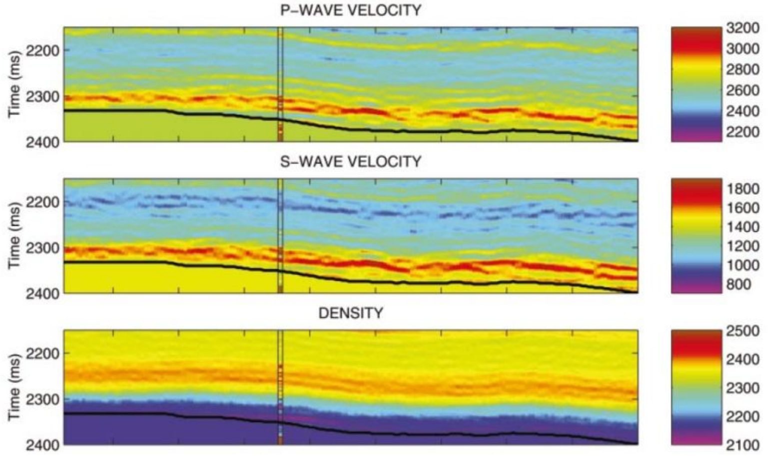
$$\mathbf{\Sigma}_d \text{ is large} \quad \Rightarrow \quad \begin{cases} \mathbf{\Sigma}_{\mathbf{m}|\mathbf{d}} \simeq \mathbf{\Sigma}_m \\ \mu_{\mathbf{m}|\mathbf{d}} \simeq \mu_m \end{cases}$$

Final result; posterior (α, β, ρ):

$$\mathbf{m} = [\ln \alpha, \ln \beta, \ln \rho]^T$$

$$\begin{aligned} \mu_{\mathbf{m}|\mathbf{d}} &= [\ln \mu_\alpha, \ln \mu_\beta, \ln \mu_\rho]^T \\ \Rightarrow \mu_{(\alpha,\beta,\rho)} &= e^{\mu_{\mathbf{m}|\mathbf{d}}} \end{aligned}$$

$$\begin{aligned} \alpha &= e^{\mu_1} \\ \beta &= e^{\mu_2} \\ \rho &= e^{\mu_3} \end{aligned}$$



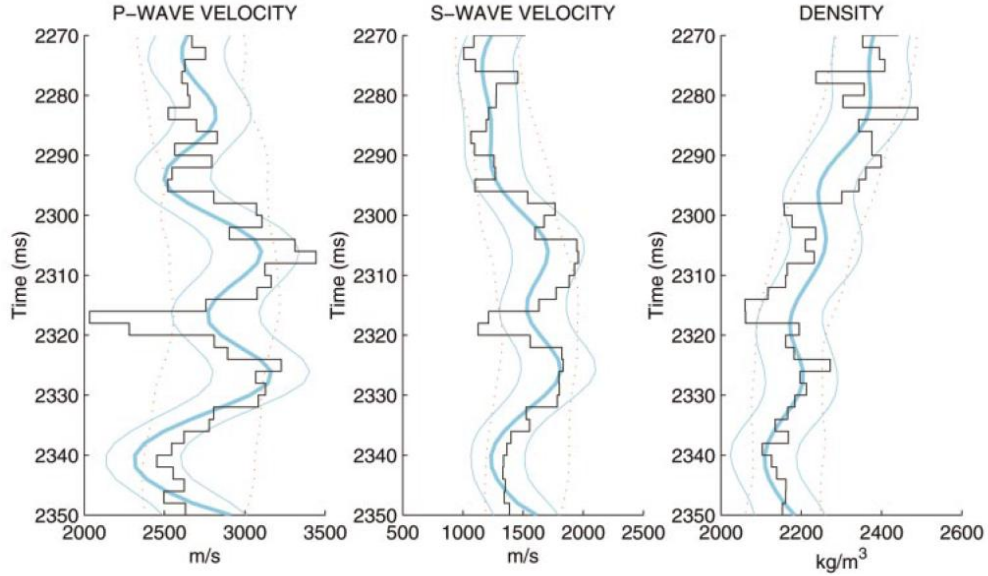


FIG. 14. The MAP solution (thick blue line) in the well position with 0.95 prediction interval (thin blue lines), the well log (black line), and 0.95 prior model interval (red dotted lines).

(Buland and Omre, Geophysics, 2003)

AVO equations (1)

Aki&Richards approximation:

$$R_{PP} = \frac{1}{2}[1+\tan^2\theta]\frac{\Delta\alpha}{\alpha} - 4\left(\frac{\beta}{\alpha}\right)^2 \sin^2\theta\frac{\Delta\beta}{\beta} + \frac{1}{2}\left[1 - 4\left(\frac{\beta}{\alpha}\right)^2 \sin^2\theta\right]\frac{\Delta\rho}{\rho}$$

3-term Fatti equation:

$$\begin{aligned} R_{PP} &= \frac{1}{2}[1+\tan^2\theta]\frac{\Delta Z_P}{Z_P} - 4\left(\frac{\beta}{\alpha}\right)^2 \sin^2\theta\frac{\Delta Z_S}{Z_S} + \frac{1}{2}\left[4\left(\frac{\beta}{\alpha}\right)^2 \sin^2\theta - \tan^2\theta\right]\frac{\Delta\rho}{\rho} \\ Z_\eta &= \rho v_\eta \end{aligned}$$

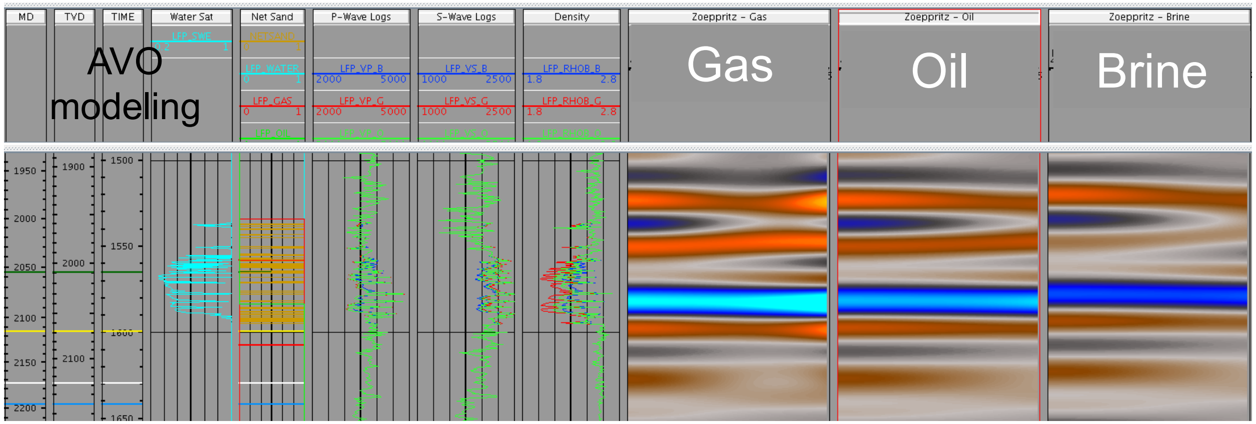
AVO equations (2)

Shuey approximation:

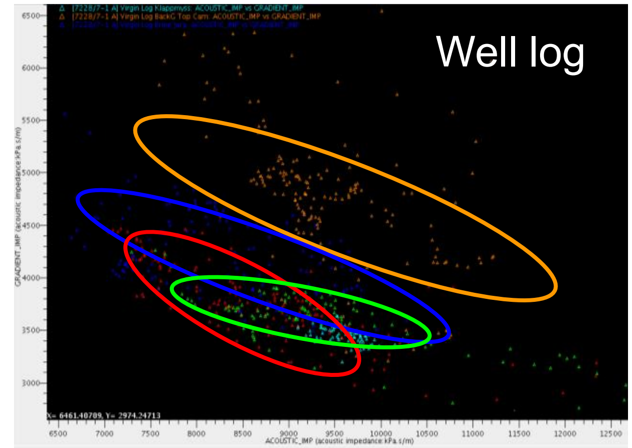
$$R_{PP}(\theta) = R_0 + G\sin^2\theta + H(\tan^2\theta - \sin^2\theta)$$

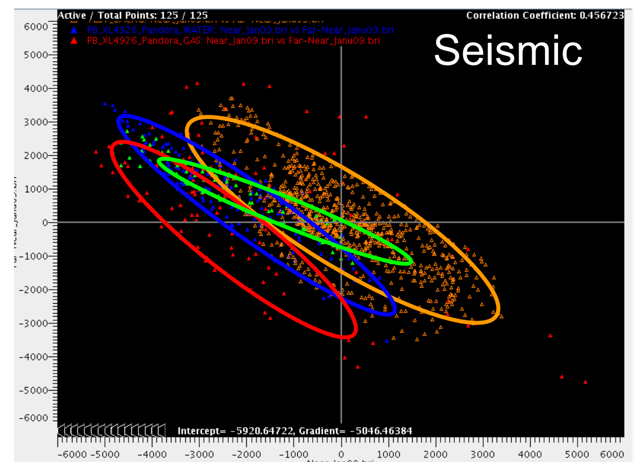
$$\begin{aligned}
R_0 &= \frac{1}{2}\left[\frac{\Delta\alpha}{\alpha} + \frac{\Delta\rho}{\rho}\right] \\
G &= \frac{1}{2}\left[\frac{\Delta\alpha}{\alpha} - 4\left(\frac{\beta}{\alpha}\right)^2\left(\frac{\Delta\rho}{\rho} + 2\frac{\Delta\beta}{\beta}\right)\right] \\
H &= \frac{1}{2}\frac{\Delta\alpha}{\alpha}
\end{aligned}$$

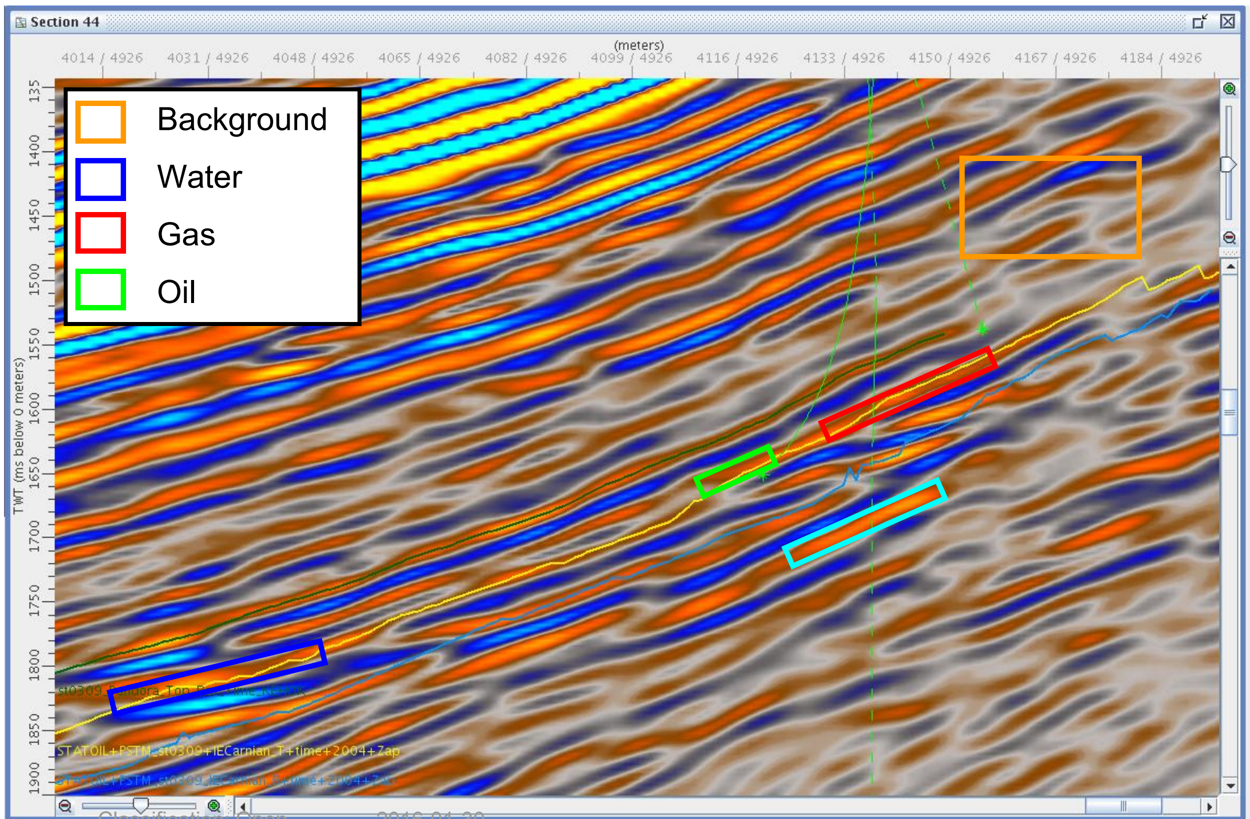
Simple analysis using the small-angle Shuey approximation



Pandora well-line







The main difference is between HC and Brine trends, not oil and gas

Least squares solution

Define objective function:

$$\Phi \quad = \quad \frac{1}{2}(\hat{\mathbf{d}} - \hat{\mathbf{A}}\mathbf{m})^T(\hat{\mathbf{d}} - \hat{\mathbf{A}}\mathbf{m})$$

Minimize $\Phi \Leftrightarrow$ maximize L:

$$\frac{\partial \Phi}{\partial \mathbf{m}} \quad = \quad -\frac{1}{2}[\hat{\mathbf{A}}^T(\hat{\mathbf{d}} - \hat{\mathbf{A}}\mathbf{m}) + (\hat{\mathbf{d}} - \hat{\mathbf{A}}\mathbf{m})^T\hat{\mathbf{A}}] = 0$$

Least-squares (MLH) solution:

$$\mathbf{m} = \underbrace{[\hat{\mathbf{A}}^T\hat{\mathbf{A}}]^{-1}\hat{\mathbf{A}}^T}_{\text{pseudo inverse}}\hat{\mathbf{d}}$$

Minimum-norm solution

Under-determined linear problem:

$$\begin{aligned} \mathbf{d} &= \mathbf{A}\mathbf{m} + \mathbf{n} \\ N_d &< N_m \end{aligned}$$

Fewer observations (equations) then unknowns

Minimum-norm:

$$\begin{aligned} &\text{minimize} && \mathbf{m}^T\mathbf{m} \\ &\text{subject to} && \mathbf{A}\mathbf{m} - \mathbf{d} = 0 \end{aligned}$$

Objective function; Lagrange multipliers

$$\Phi = \mathbf{m}^T\mathbf{m} + \lambda^T(\mathbf{A}\mathbf{m} - \mathbf{d})$$

Minimize Φ w.r.t. $\mathbf{m}$ and λ:

$$\frac{\partial \Phi}{\partial \mathbf{m}} = 0$$
$$\frac{\partial \Phi}{\partial \lambda} = 0$$

Hence

$$\begin{aligned} 2\mathbf{m} + \mathbf{A}^T\lambda &= 0 \\ \mathbf{A}\mathbf{m} - \mathbf{d} &= 0 \end{aligned}$$

From first equation

$$\mathbf{m} = -\frac{1}{2}\mathbf{A}^T\lambda$$

Substitute in 2nd equation

$$\frac{1}{2}\mathbf{A}\mathbf{A}^T\lambda + \mathbf{d} = 0$$

$$\lambda = -2(\mathbf{A}\mathbf{A}^T)^{-1}\mathbf{d}$$

Backsubstitute to eliminate λ

$$\mathbf{m} = \mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1}\mathbf{d}$$

Least squares vs. minimum norm

Least-squares solution

$$\begin{aligned} \mathbf{m} &= (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{d} \\ N_d &\geq N_m \end{aligned}$$

- Over-determined linear problems
- More equations than unknowns

Minimum-norm solution

$$\begin{aligned} \mathbf{m} &= \mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1}\mathbf{d} \\ N_d &\leq N_m \end{aligned}$$

- Under-determined linear problems
- Less equations than unknowns

Moore-Penrose pseudo inverses